

Medical Vision-Language Models:

From pre-training to finetuning and adaptation

MICCAI 2026 Tutorial on Multimodal Foundation Models for Biomedical Imaging
(MULTIFOUND)

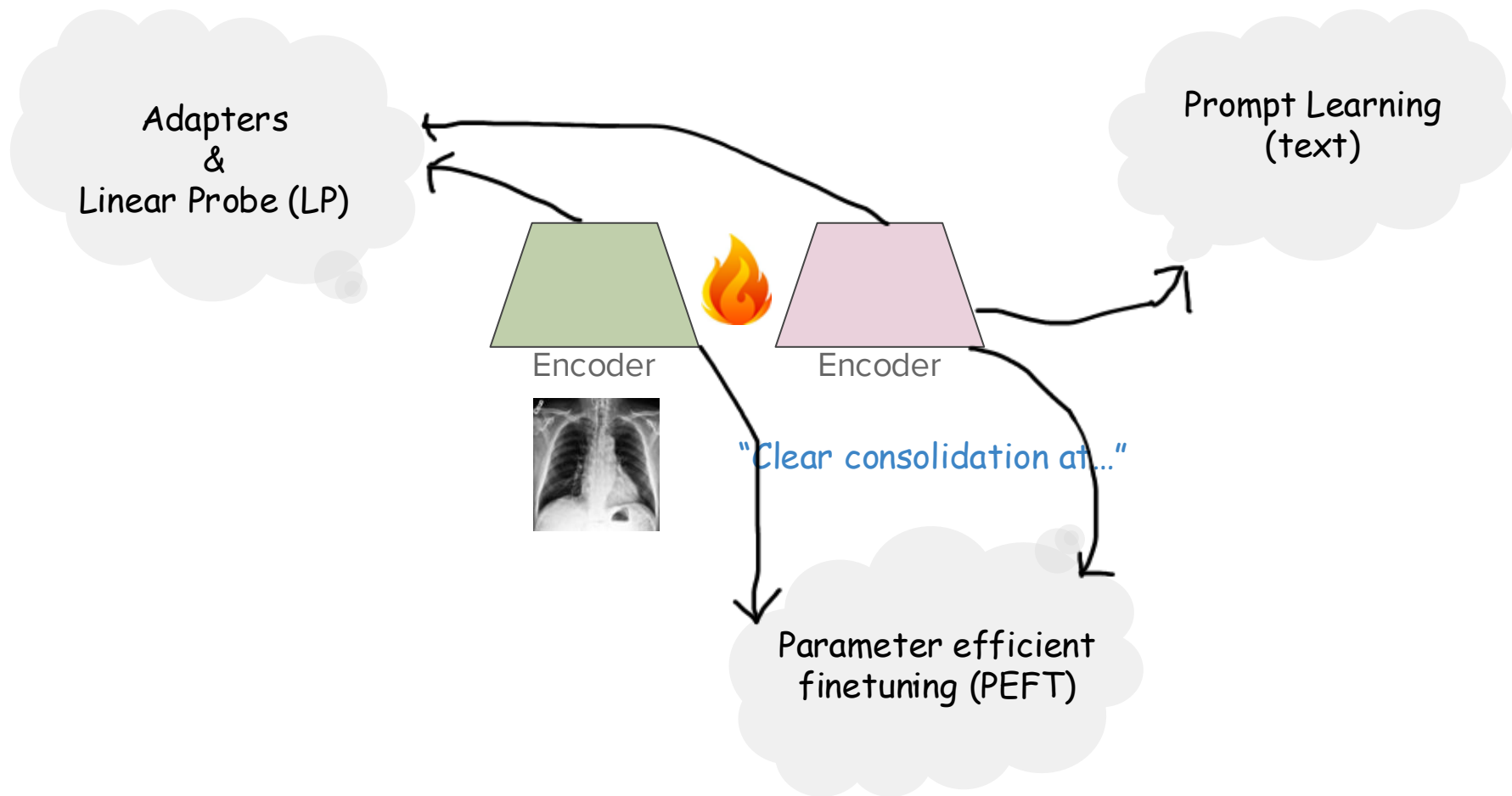
Parts 3

Ismail Ben Ayed

Part 3

Fine-tuning and Adaptation of Foundation Models

Fine-tuning VLMs in downstream tasks?

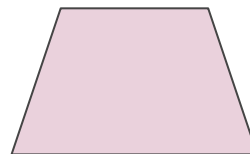


Important considerations (in practical scenarios, e.g., clinical settings)?

Limited computation
&
memory resources



Encoder



Encoder

"Clear consolidation at..."

Few-shot regimes

Black box
adaptation

What and how to fine-tune?

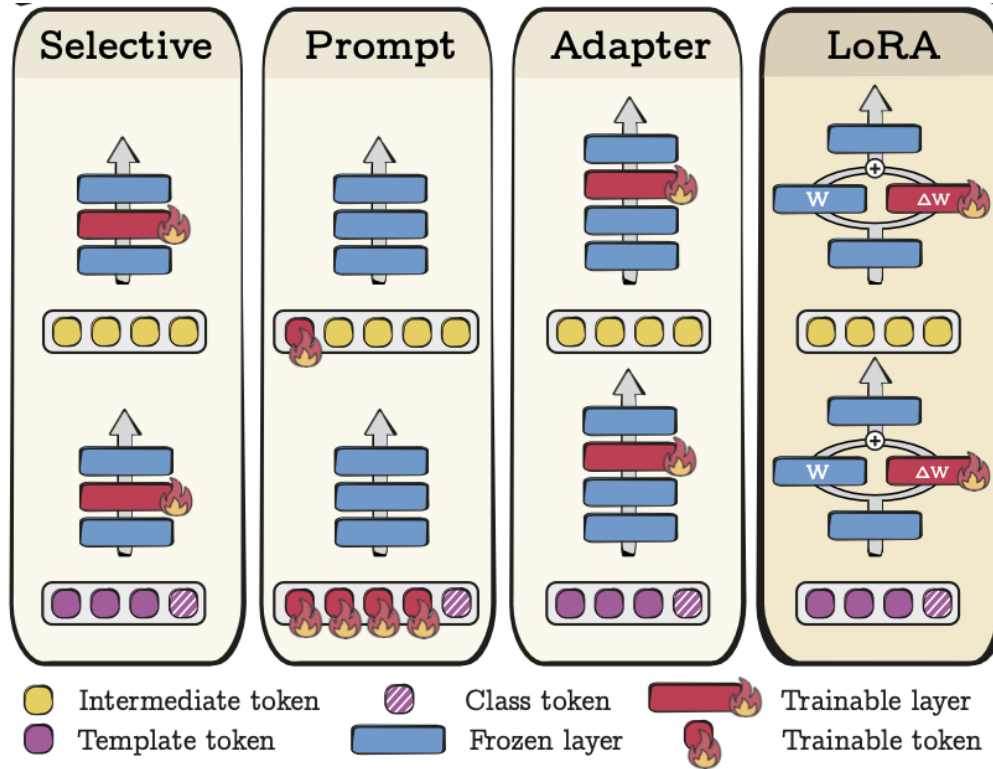
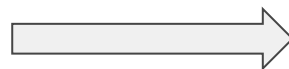
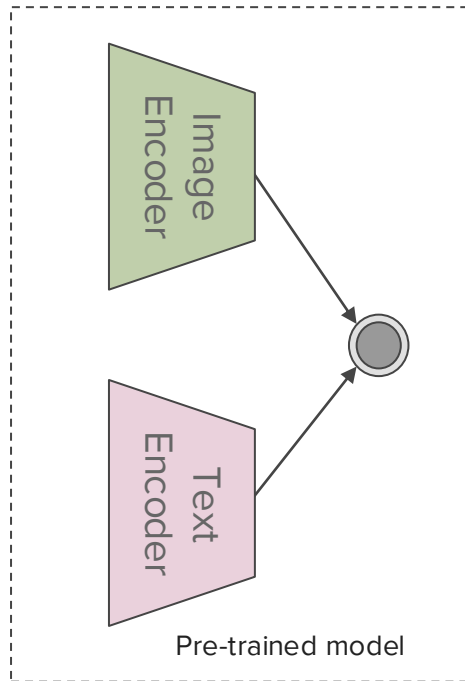


Figure from: Zanella & Ben Ayed et al., Low-Rank Few-Shot Adaptation of Vision-Language Models, CVPRW 2024

Adaptation in few-shot regimes



Adapt it
with a few
labelled samples

Class 1



Class 2

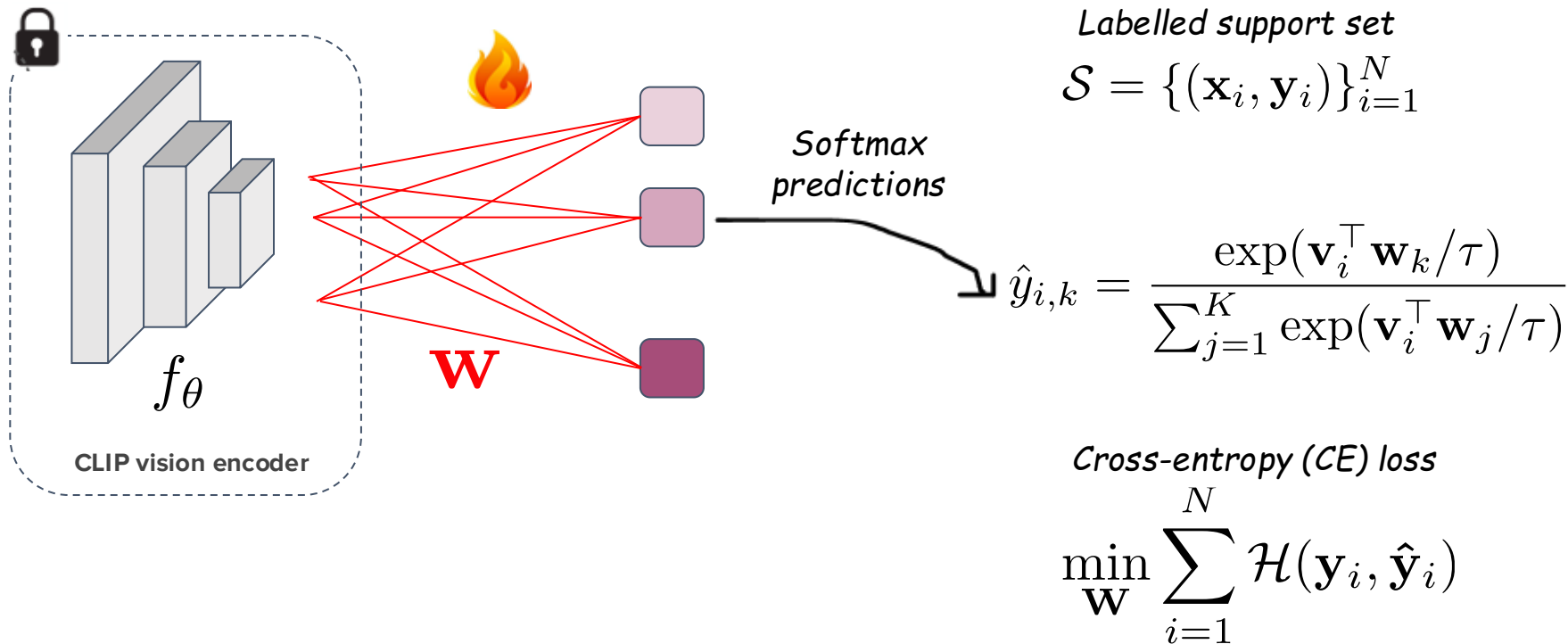


Class n

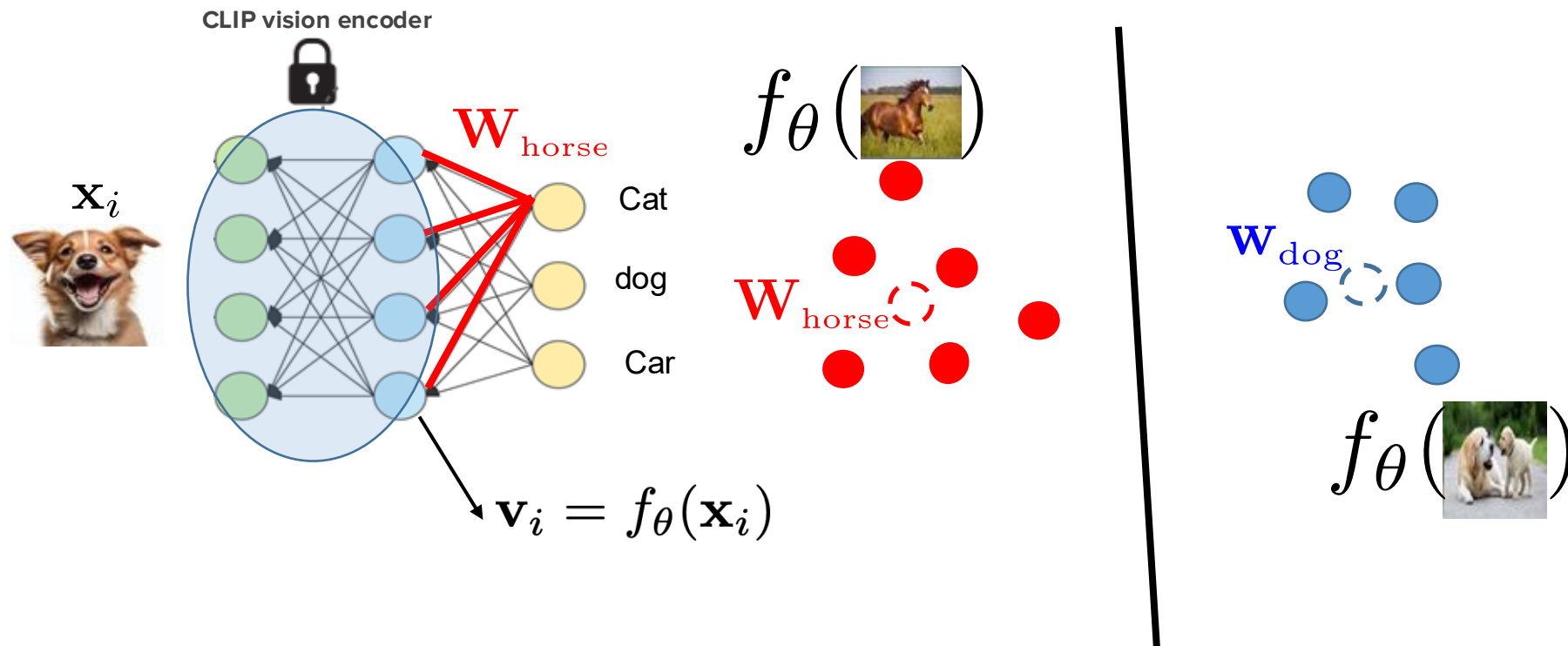


$$\mathcal{S} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$$

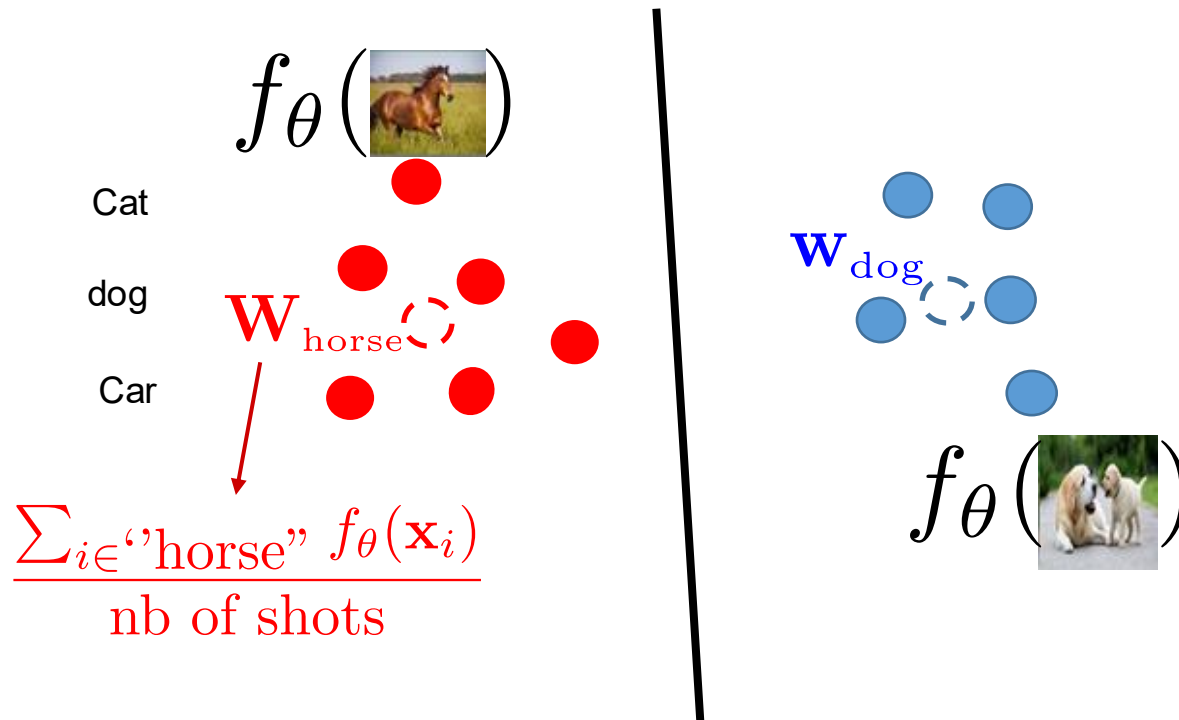
Preliminaries: What is Linear Probing (LP)?



LP: Linear-classifier weights are (almost) class means!



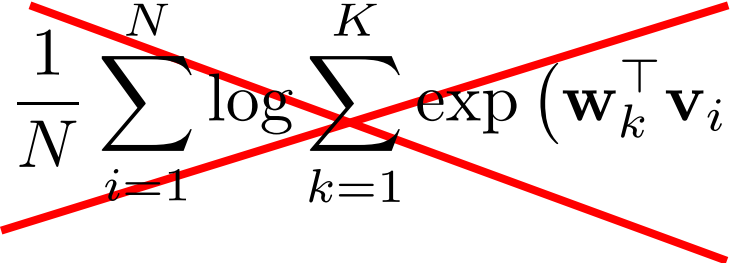
“Training-free” Version: SimpleShot (almost a LP)



“Training-free” Version: SimpleShot (almost a LP)

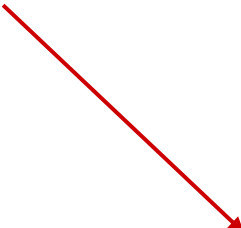
$$\text{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_k y_{i,k} \mathbf{w}_k^\top \mathbf{v}_i + \frac{1}{N} \sum_{i=1}^N \log \sum_{k=1}^K \exp(\mathbf{w}_k^\top \mathbf{v}_i)$$

“Training-free” Version: SimpleShot (almost a LP)

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“Training-free” Version: SimpleShot (almost a LP)

$$\min_{\mathbf{w}} \text{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_k y_{i,k} \mathbf{w}_k^\top \mathbf{z}_i + \frac{1}{2} \sum_{k=1}^K \|\mathbf{w}_k\|^2$$



What is the closed-form solution here?

Learning to prompt: Generating context-specific text

CoOp

Caltech101



Prompt	Accuracy
a [CLASS].	82.68
a photo of [CLASS].	80.81
a photo of a [CLASS].	86.29
$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	91.83

Flowers102



Prompt	Accuracy
a photo of a [CLASS].	60.86
a flower photo of a [CLASS].	65.81
a photo of a [CLASS], a type of flower.	66.14
$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	94.51

Learning to prompt: Generating context-specific text

CoOp

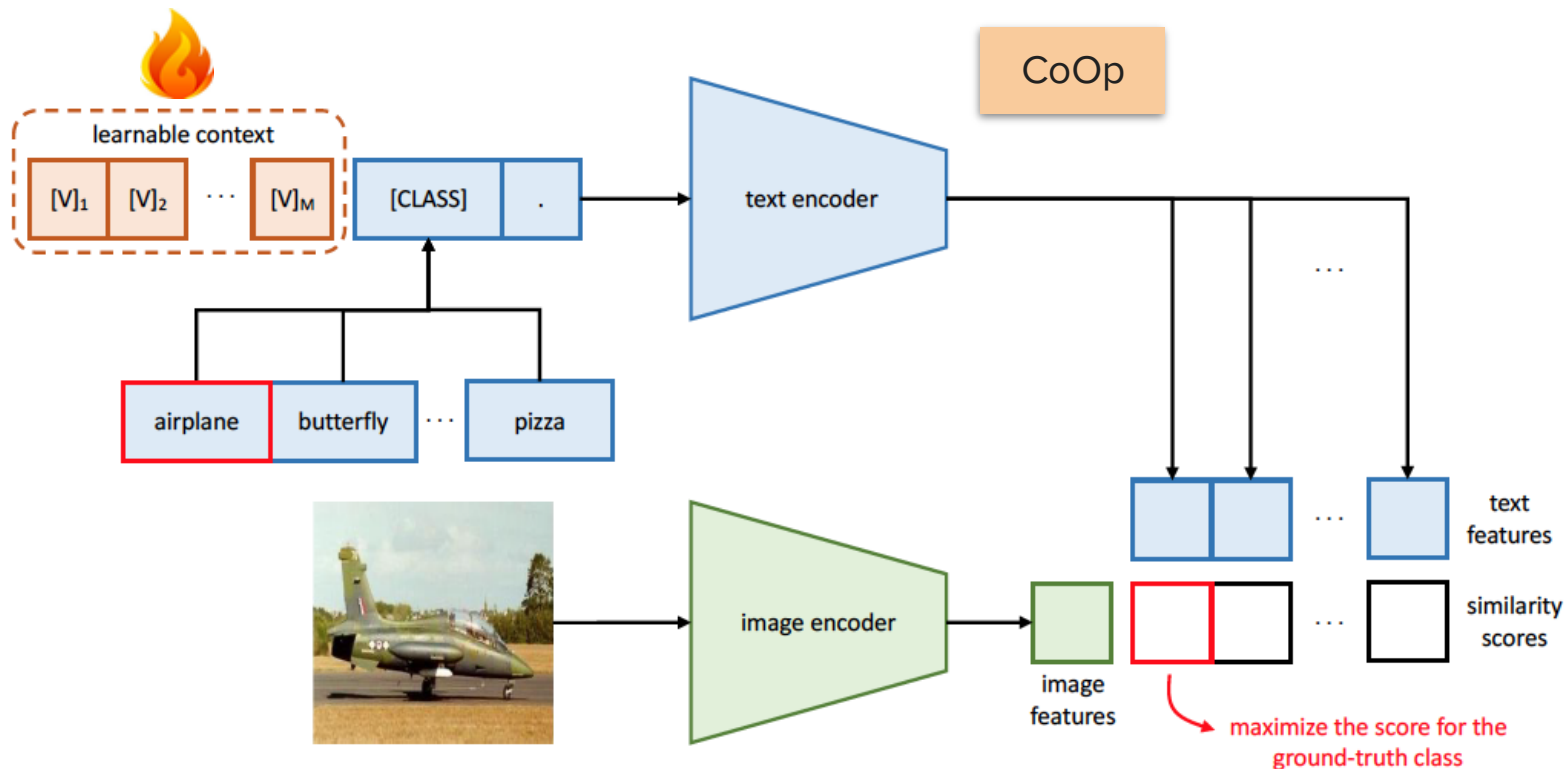
Caltech101	Prompt	Accuracy
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~3,000 citations
&

A crowded CVPR Workshop on Prompt Learning!

Learning to prompt: Generating context-specific text



Learning to prompt: Generating context-specific text

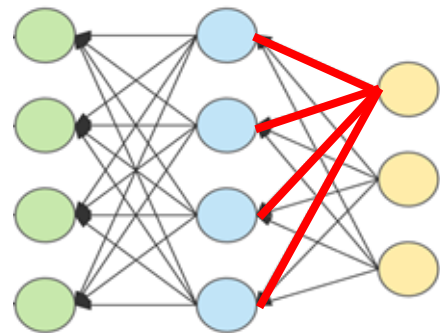
Prompt learning (min CE)

$$g_{\phi}([V]_1, \dots, [V]_m, [\text{DOG}])$$

$$f_{\theta}(\text{image of a horse})$$

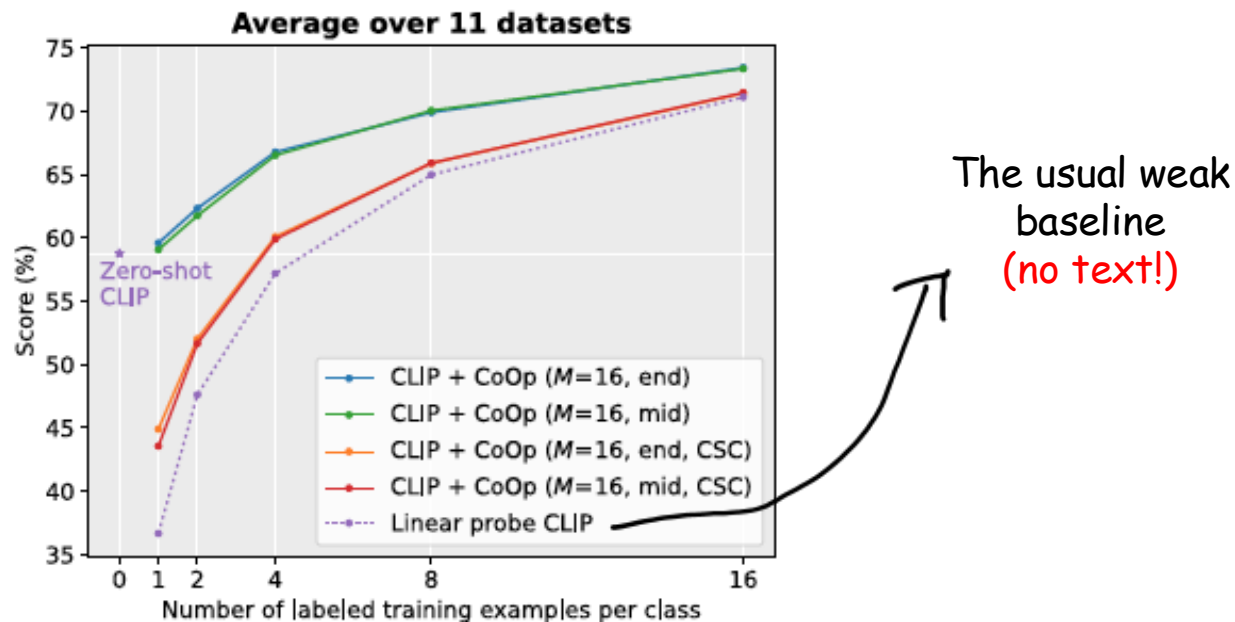
W_{horse}

Standard LP (min CE)

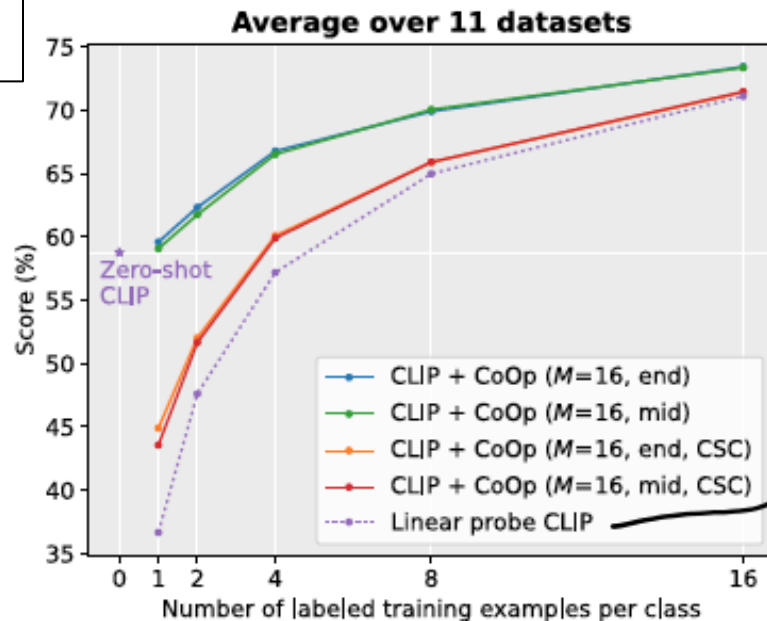
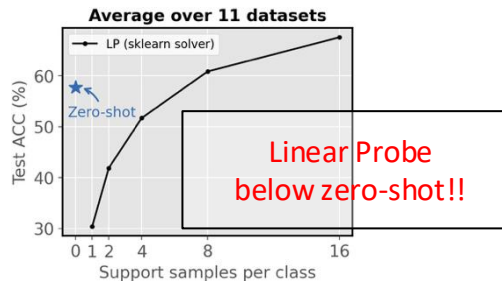


W_{horse}

Prompt learning



Prompt learning



The usual weak
baseline
(no text!)

CoOp: Zhou et al., Learning to Prompt for Vision-Language Models, IJCV 2022[~3000 citations]

CoCoOp: Zhou et al., Conditional Prompt Learning for Vision-Language Models, CVPR 2022[~1200 citations]

KgCoOp: Yao et al., Prompt Tuning with Knowledge-guided Context Optimization, CVPR 2023[~122 citations]

Prompt learning

- Heavy computational load
 - backpropagate through the text encoder
- Not valid in black-box scenario

CoOp: Zhou et al., Learning to Prompt for Vision-Language Models, IJCV 2022[~3000 citations]

CoCoOp: Zhou et al., Conditional Prompt Learning for Vision-Language Models, CVPR 2022[~1200 citations]

KgCoOp: Yao et al., Prompt Tuning with Knowledge-guided Context Optimization, CVPR 2023[~122 citations]

Prompt learning: What is it doing better than LP (intuitively)?

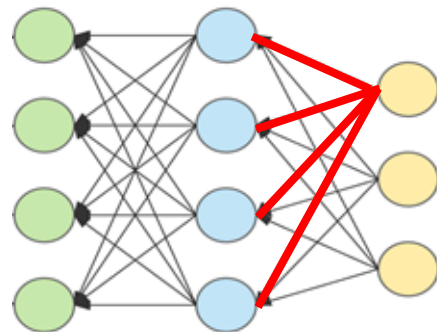
Prompt learning (min CE)

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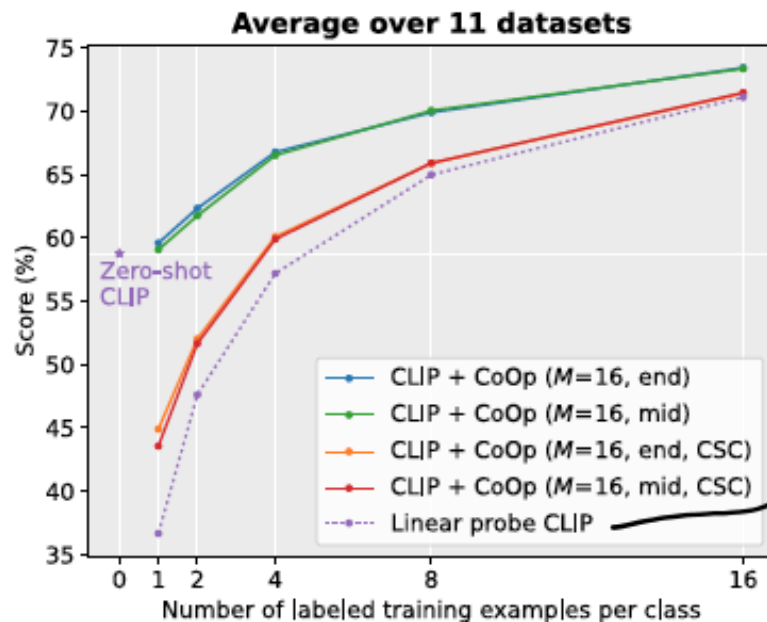
W_{horse}

Standard linear probe (min CE)



W_{horse}

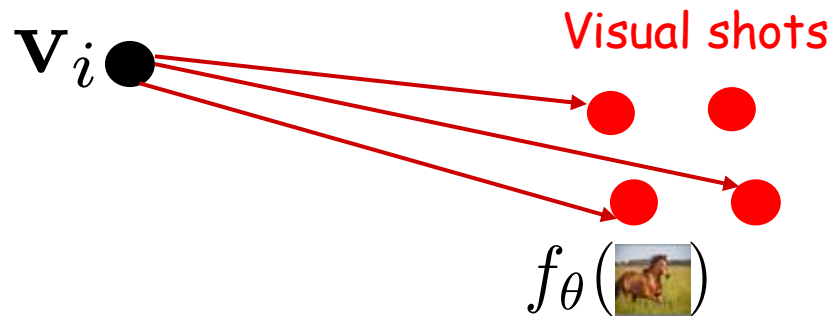
Prompt learning



The usual weak
baseline
(no text encoder!)

An example of adapter (Tip-Adapter): Blending image and text features

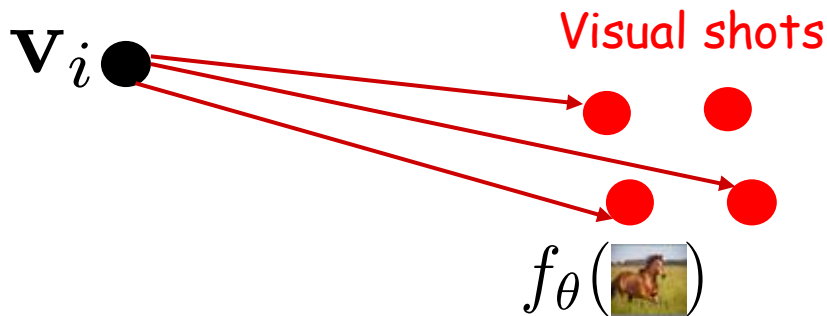
$$\text{score}_{i, \text{horse}} \propto \exp \sum_j \mathbf{v}_i^\top \mathbf{v}_j$$



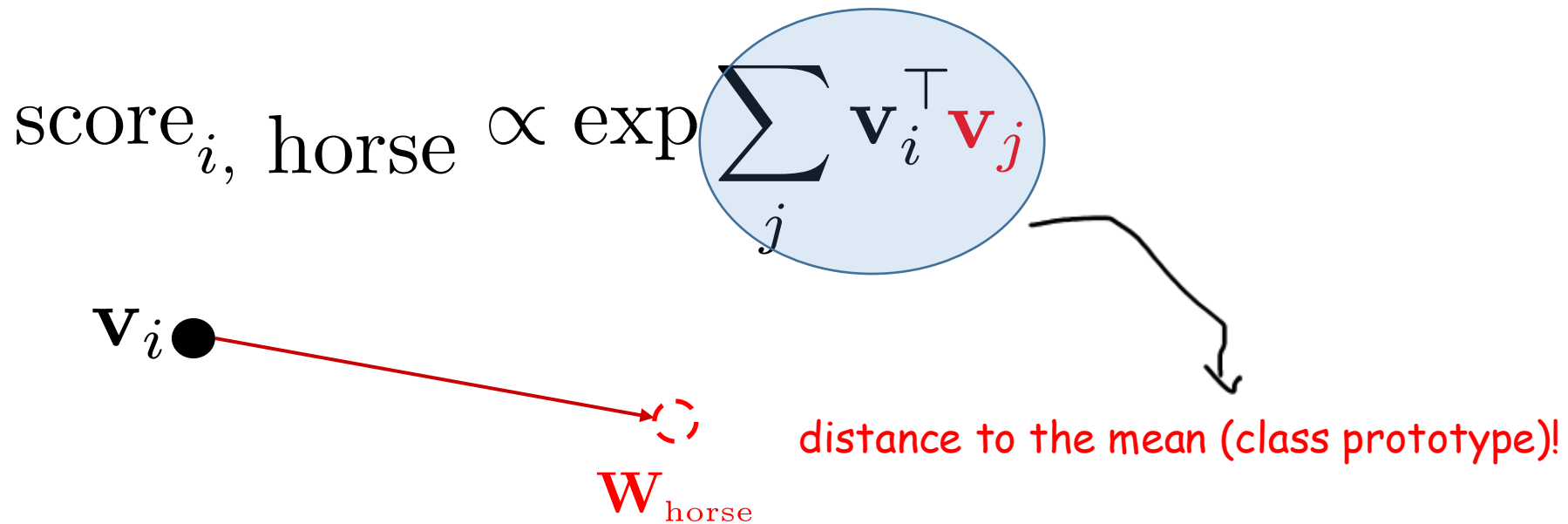
An example of adapter (Tip-Adapter): Blending image and text features

Almost linear in $[0,1]$

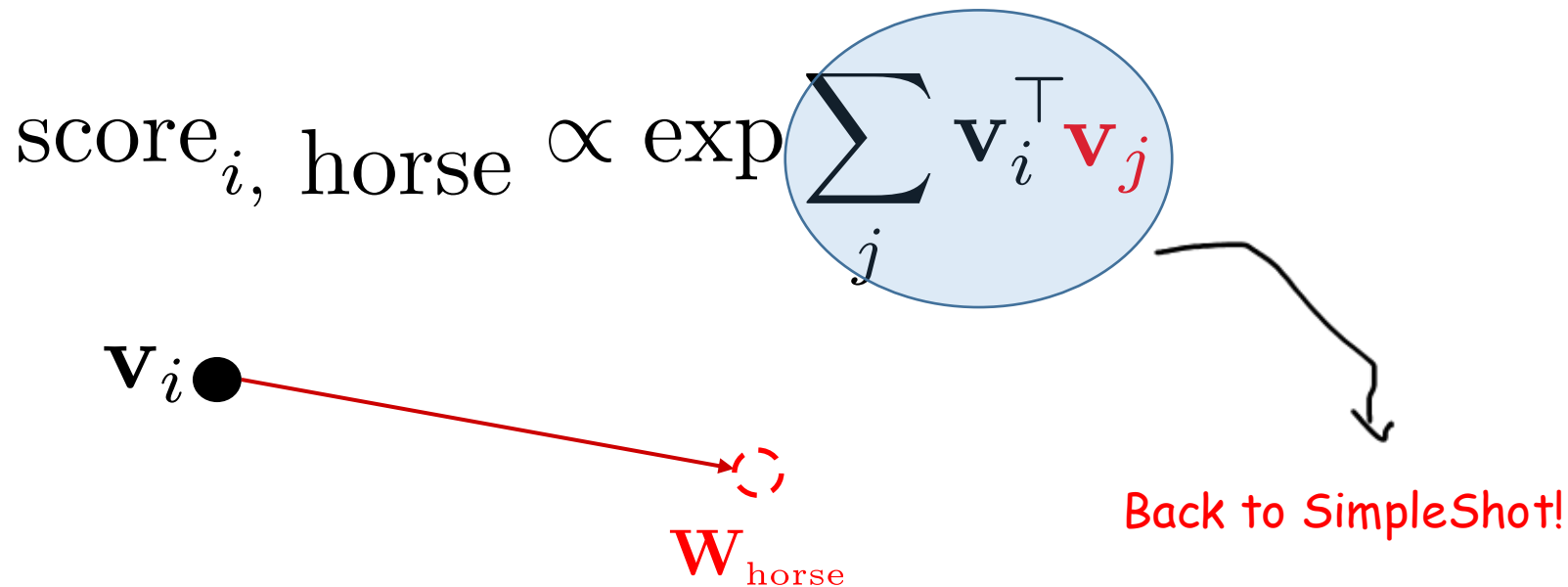
$$\text{score}_{i, \text{horse}} \propto \exp \sum_j \mathbf{v}_i^\top \mathbf{v}_j$$



An example of adapter (Tip-Adapter): Blending image and text features



An example of adapter (Tip-Adapter): Blending image and text features



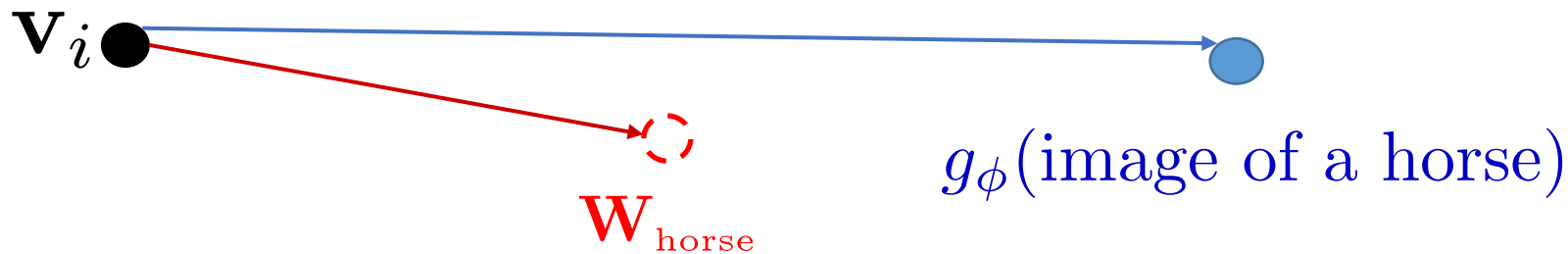
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$$\text{score}_{i, \text{horse}} \propto \exp \sum_j \mathbf{v}_i^\top \mathbf{v}_j$$



An example of adapter (Tip-Adapter): Blending image and text features

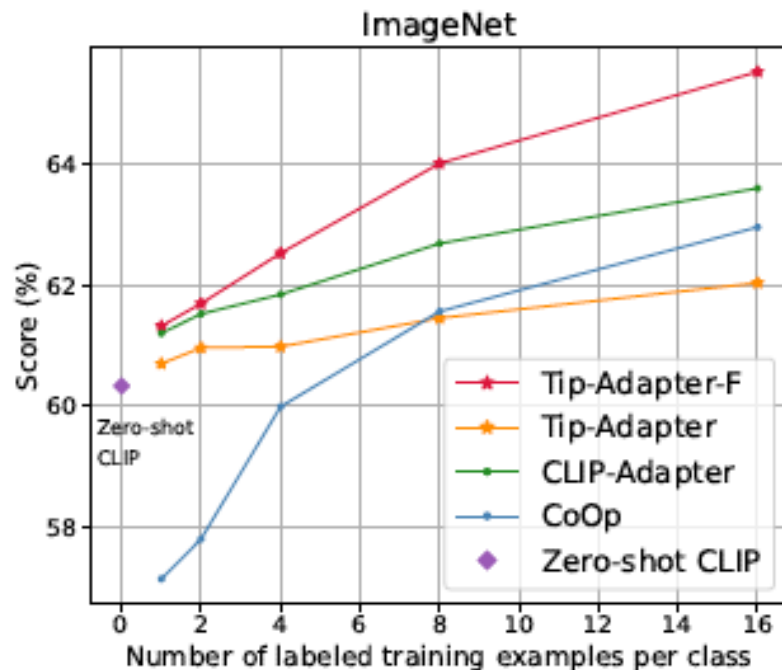
$$\text{score}_{i, \text{horse}} \propto \exp \sum_j \mathbf{v}_i^\top \{ \mathbf{v}_j + \lambda g_\phi(\text{image of a horse}) \}$$



An example of adapter (Tip-Adapter): Blending image and text features

Models	Training	Epochs	Time	Accuracy	Gain	Infer. Speed	GPU Mem.
Zero-shot CLIP [48]	Free	0	0	60.33	0	10.22ms	2227MiB
Linear-probe CLIP [48]	Required	-	13min	56.13	-4.20	-	-
CoOp [74]	Required	200	14h 40min	62.95	+2.62	299.64ms	7193MiB
CLIP-Adapter [16]	Required	200	50min	63.59	+3.26	10.59ms	2227MiB
Tip-Adapter	Free	0	0	62.03	+1.70	10.42ms	2227MiB
Tip-Adapter-F	Required	20	5min	65.51	+5.18	10.53ms	2227MiB

An example of adapter (Tip-Adapter): Blending image and text features



A closer look at adapters reveals serious flaws!!!

$$\text{score}_{i, \text{ horse}} \propto \exp \sum_j \mathbf{v}_i^\top \{ \mathbf{v}_j + \lambda g_\phi(\text{image of a horse}) \}$$

How to choose this crucial hyper-parameter?

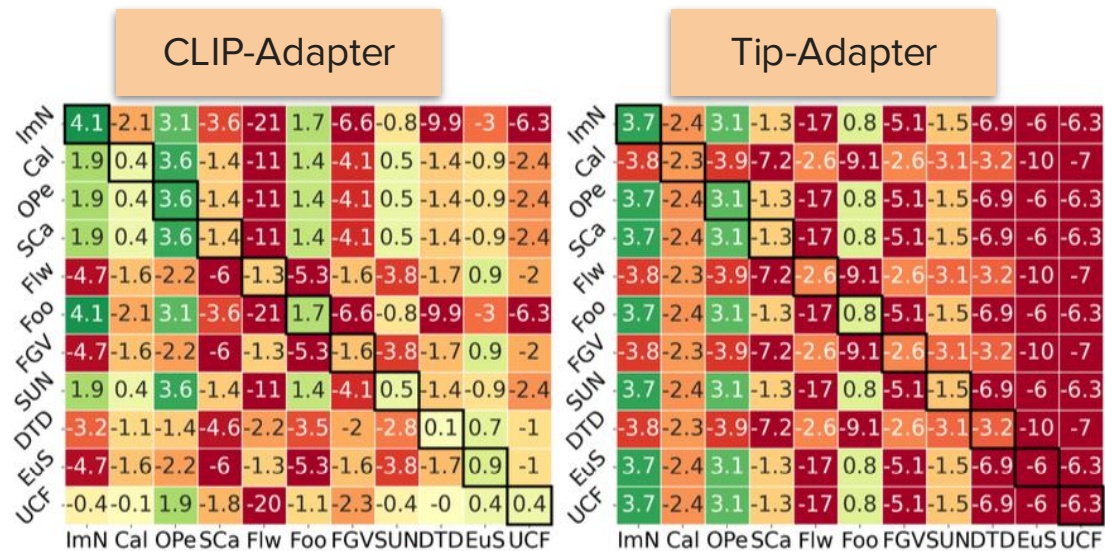
A closer look at adapters reveals serious flaws!!!

$$\text{score}_{i, \text{ horse}} \propto \exp \sum_j \mathbf{v}_i^\top \left\{ \mathbf{v}_j + \lambda g_\phi(\text{image of a horse}) \right\}$$

Don't ask me please!



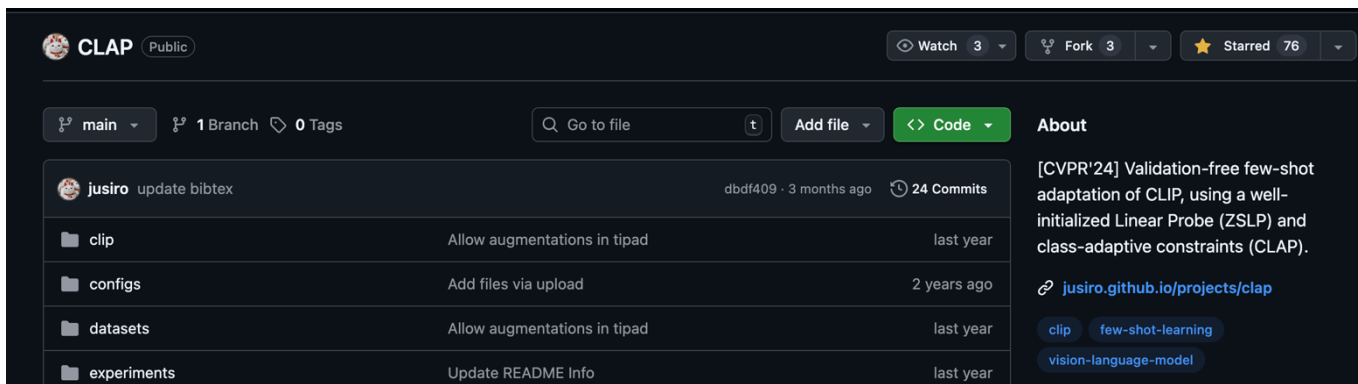
A closer look at adapters reveals serious flaws!!!



Improvement w.r.t a well-initialized linear probe!!!!

Code for this evaluation and a simple baseline
are publicly available

<https://github.com/jusiro/CLAP>



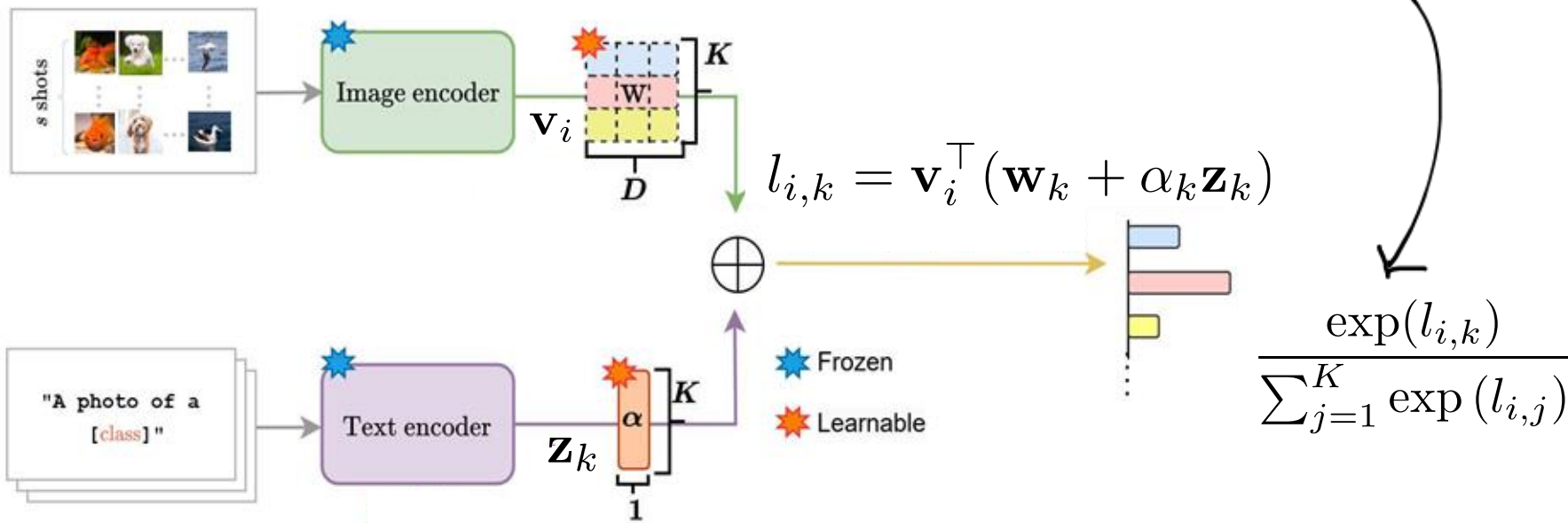
Silva-Rodriguez et al., A closer look at the few-shot adaptation of vision-language models, **CVPR 2024**

Adapters: A closer look also reveals a serious flaw!

- Depend on several key hyperparameters:
→ including the optimizer ones: LR, # of epoch, batch size, etc.
- Computationally heavy validation searches (undisclosed)
→ grid search for many hyper-parameters (some found on test data!!!)

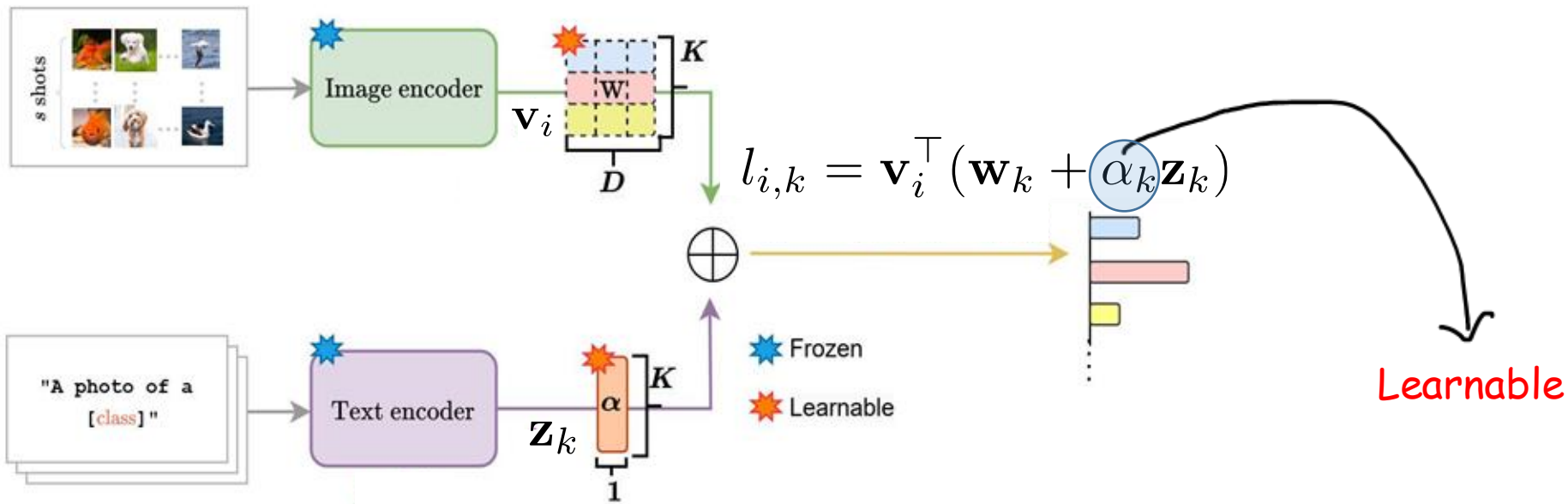
LP++: A surprisingly strong text-informed LP

$$L(\mathbf{W}, \boldsymbol{\alpha}) = -\frac{1}{N} \sum_{i=1}^N \sum_k y_{i,k} \log \hat{y}_{i,k}(\mathbf{W}, \boldsymbol{\alpha})$$

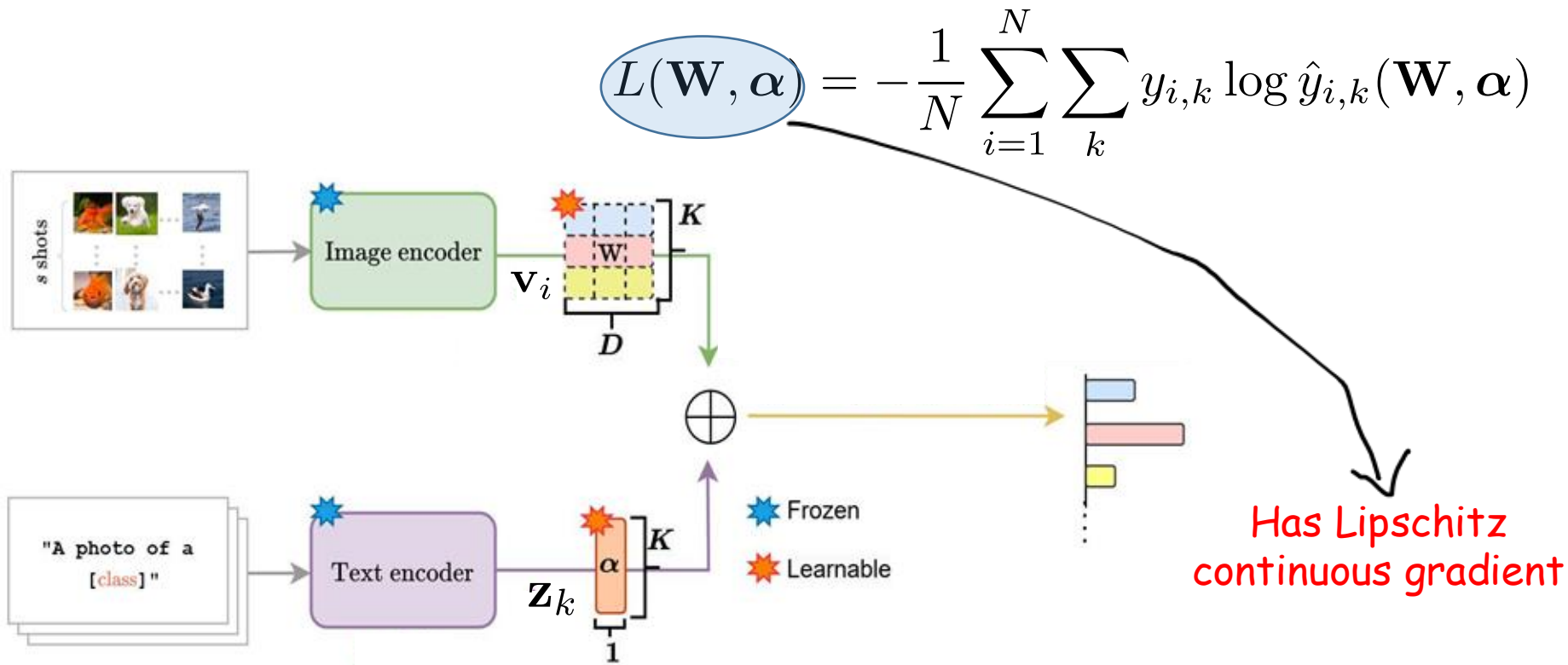


LP++: A surprisingly strong text-informed LP

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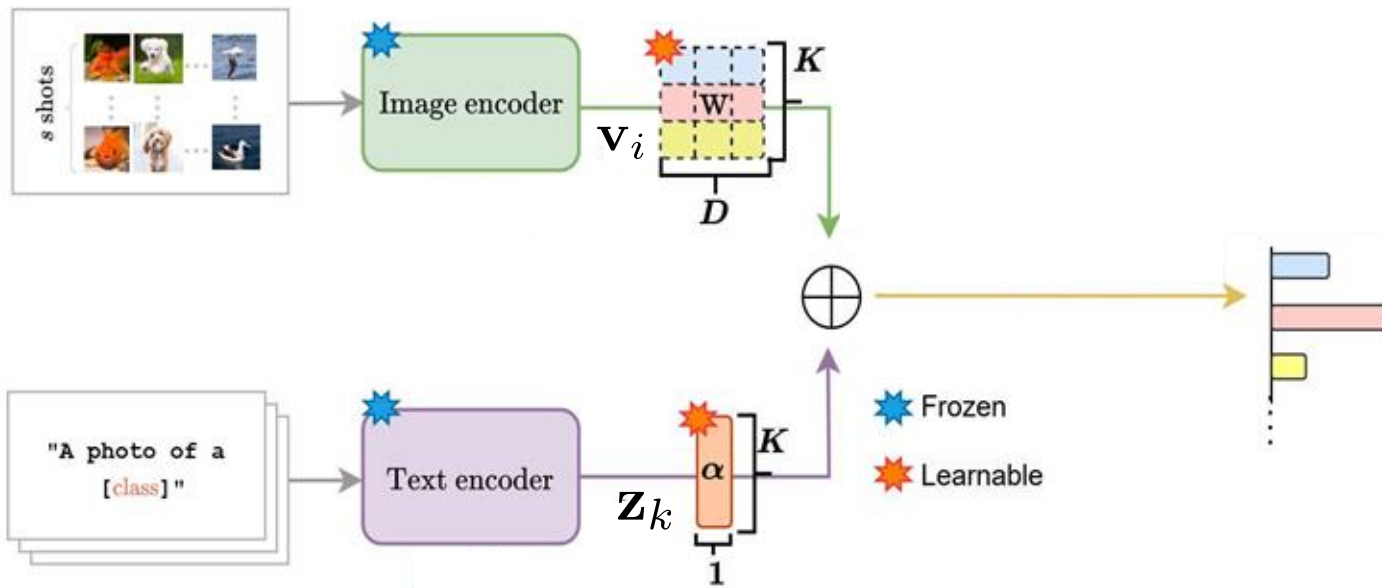


Optimization insights yield a fast, hyperparameter-free LP

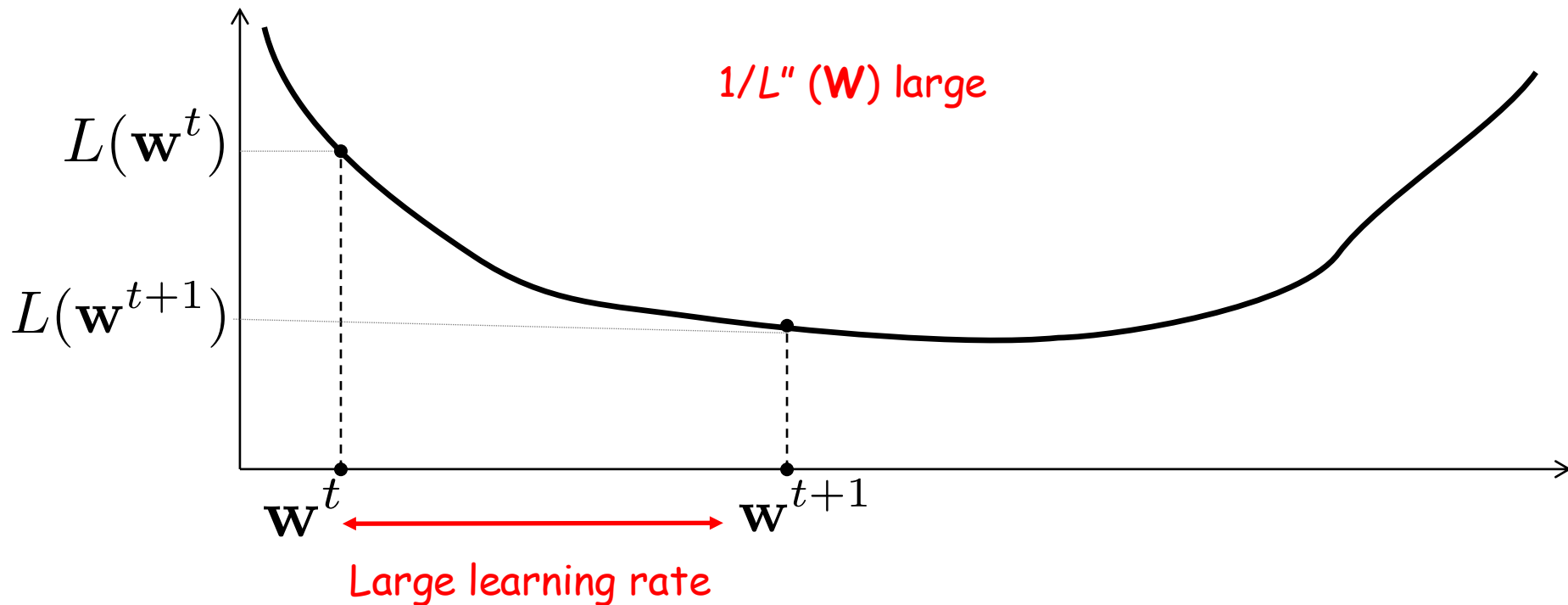


Learning rates from a bound on the Hessian max eigen value & full batch gradient descent

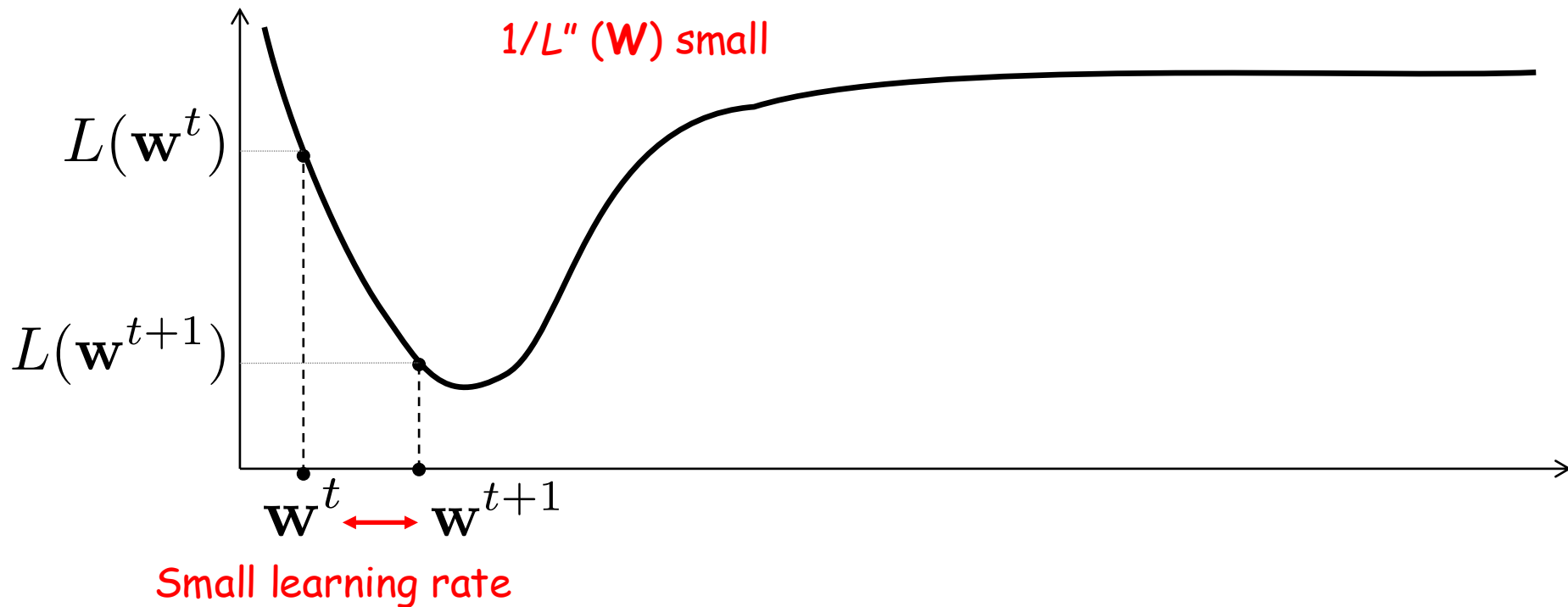
No optimization
hyper-parameters!



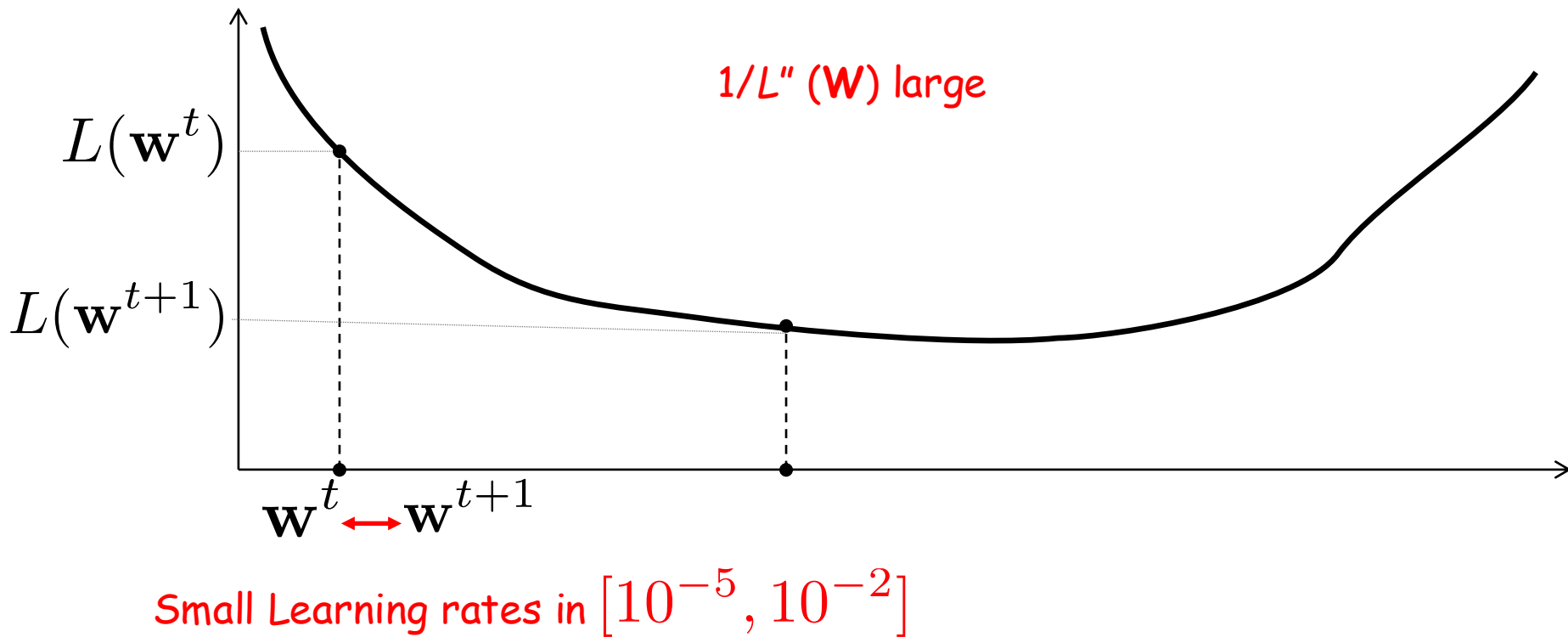
A few basic well-known aspects in convex optimization



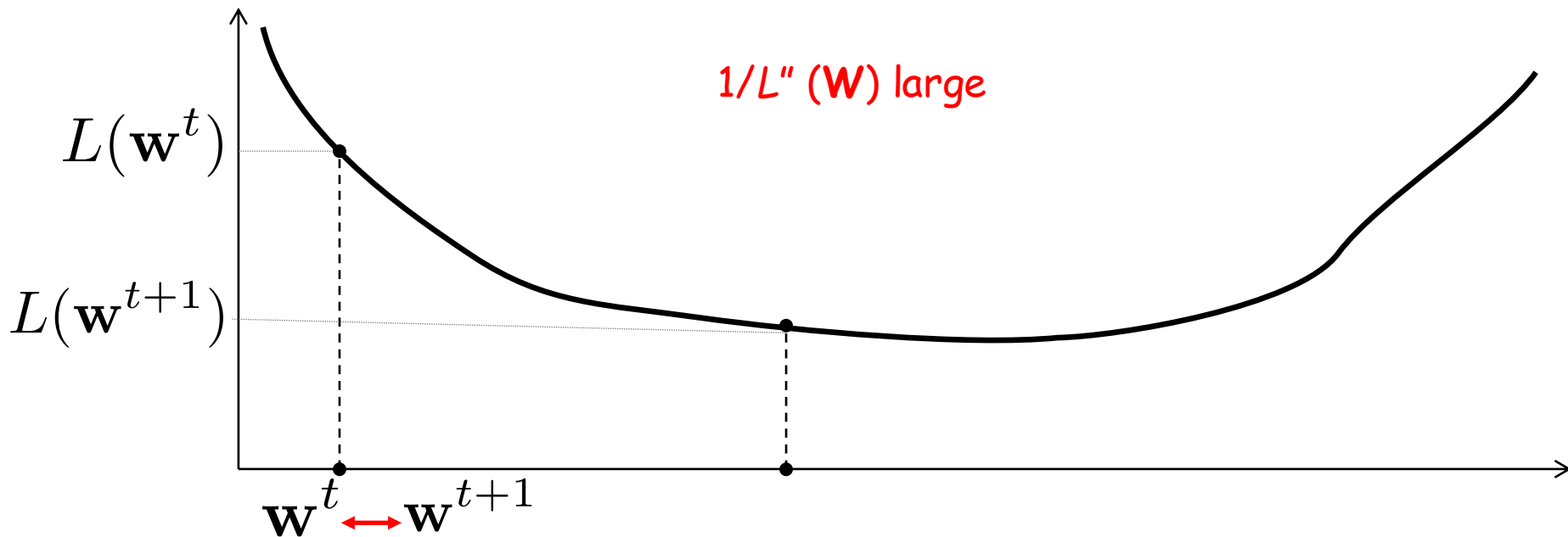
A few basic well-known aspects in convex optimization



And this is how we do it with our deep learning habits!

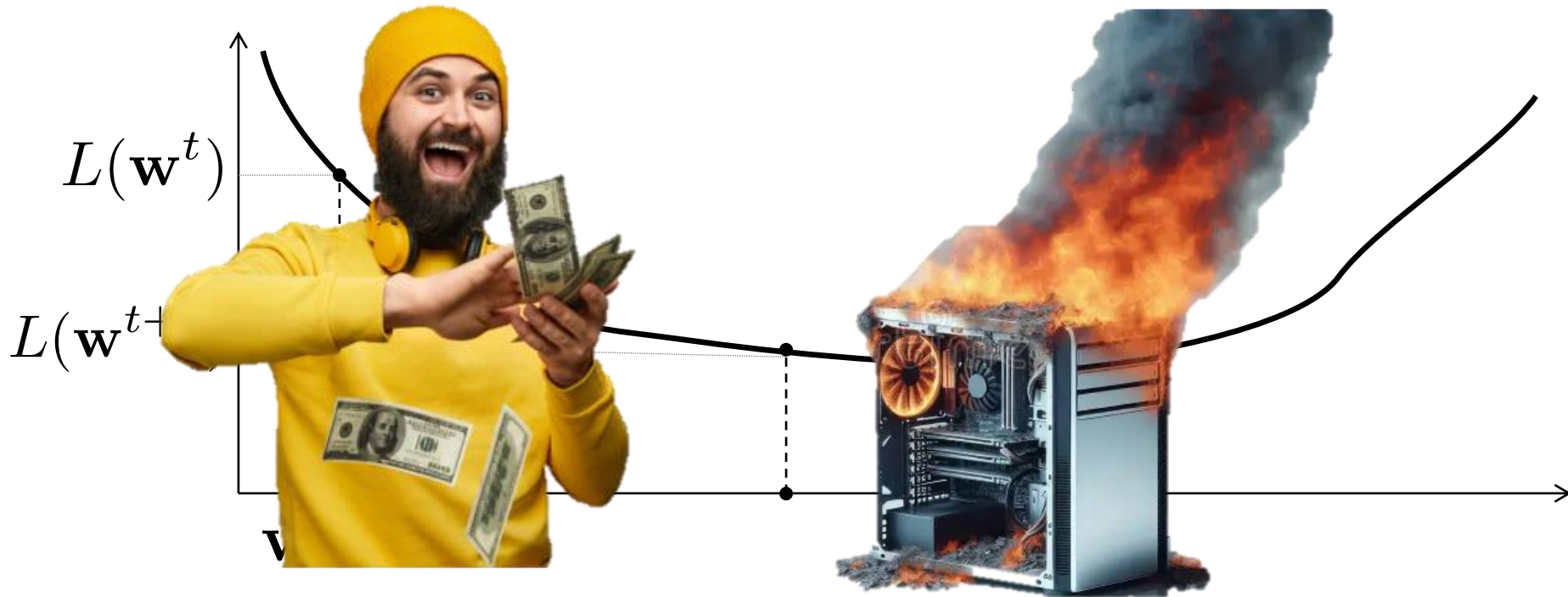


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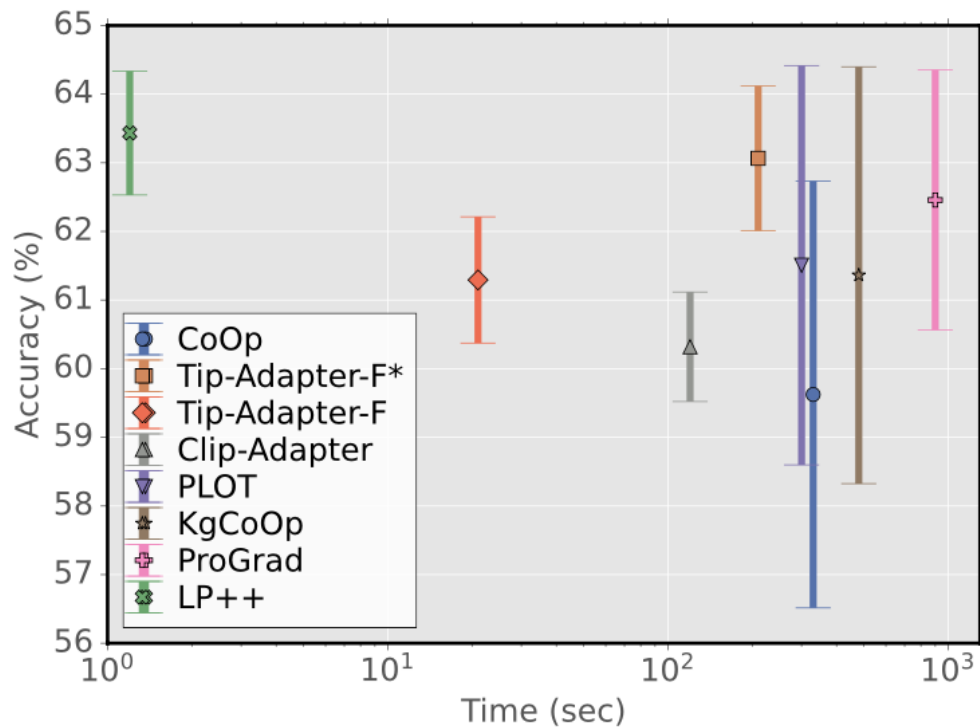


10,000 iter (run many times) + grid search over optim. hyper-parameters

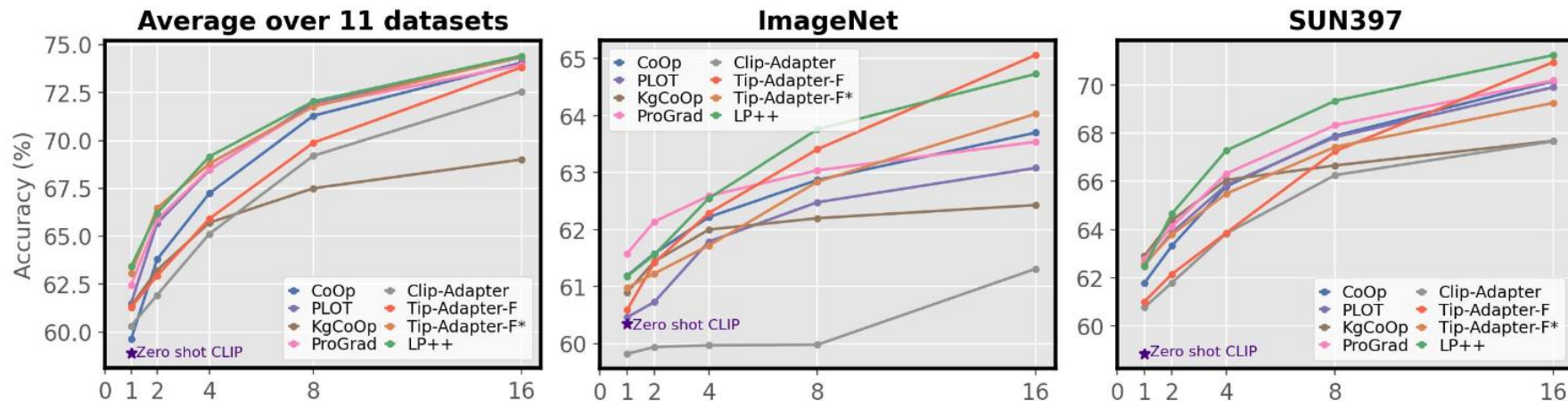
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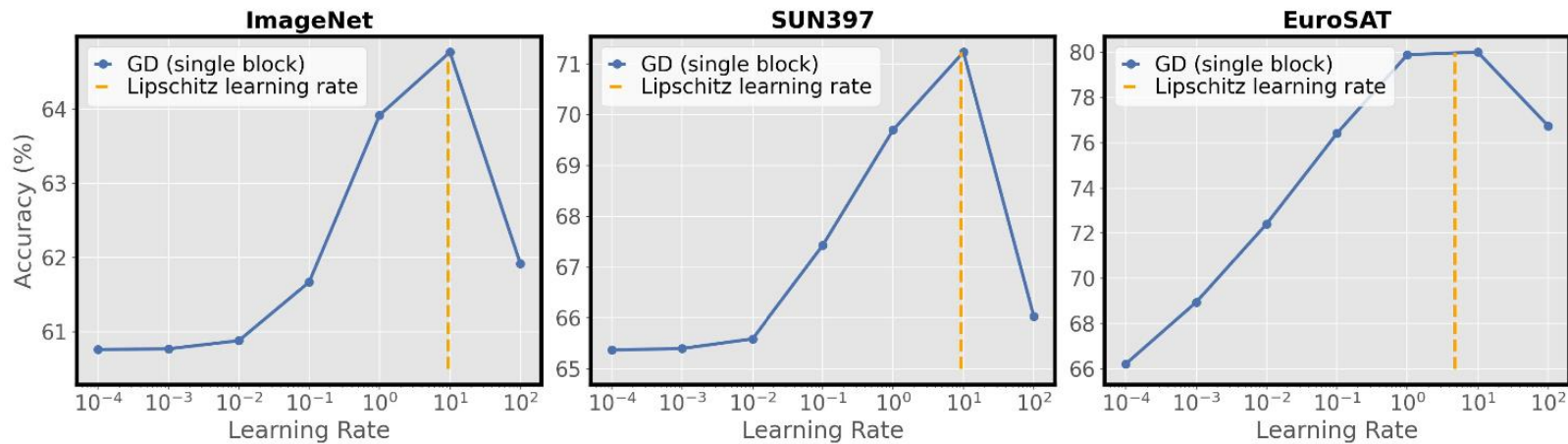
LP++: A hyper-parameter free generalization of LP



LP++: A hyper-parameter free generalization of LP



LP++: A hyper-parameter free generalization of LP



LP++: A hyper-parameter free generalization of LP

Methods	Overall Time	BlackBox	# Parameters
CoOp[29]	$\sim 17\text{h}$	$\times$	$K \times M \times D$
PLOT-2[4]	$\sim 10\text{h}$	$\times$	$P \times K \times M \times D$
KgCoOp[27]	$\sim 4\text{h}$	$\times$	$K \times M \times D$
ProGrad[30]	$\sim 20\text{h}$	$\times$	$K \times M \times D$
Clip-Adapter[9]	$\sim 40\text{min}$	$\checkmark$	$2(D_1 \times D)$
Tip-adapter-F[28]	$\sim 6\text{min}$	$\checkmark$	$K \times S \times D$
Tip-adapter-F*[28]	$\sim 50\text{min}$	$\checkmark$	$K \times S \times D$
Standard LP[23]	3min	$\checkmark$	$K \times D$
LP++	$\sim 2\text{s}$	$\checkmark$	$K(D + 1)$

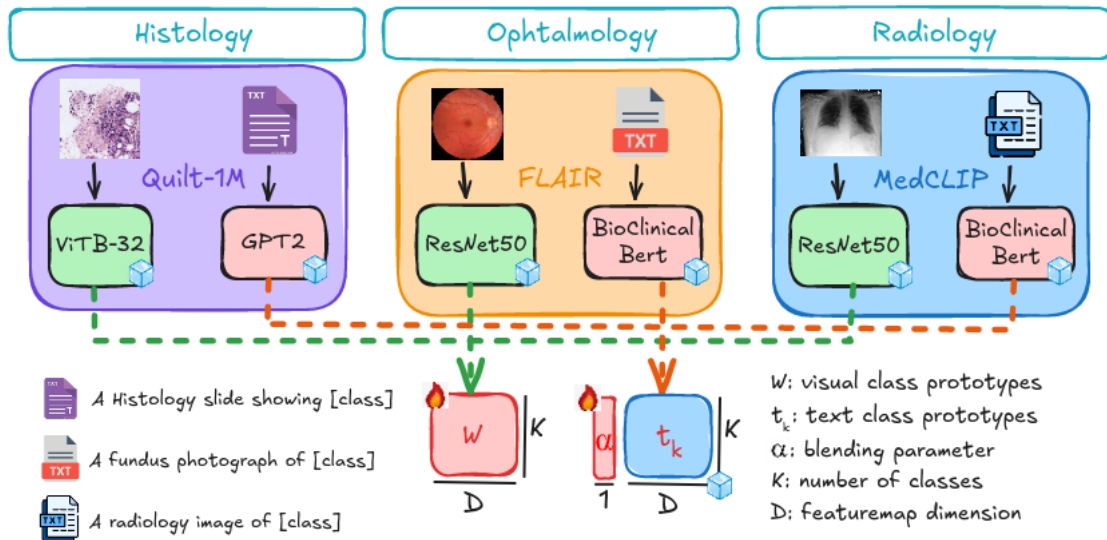
Code publicly available

<https://github.com/FereshteShakeri/FewShot-CLIP-Strong-Baseline>

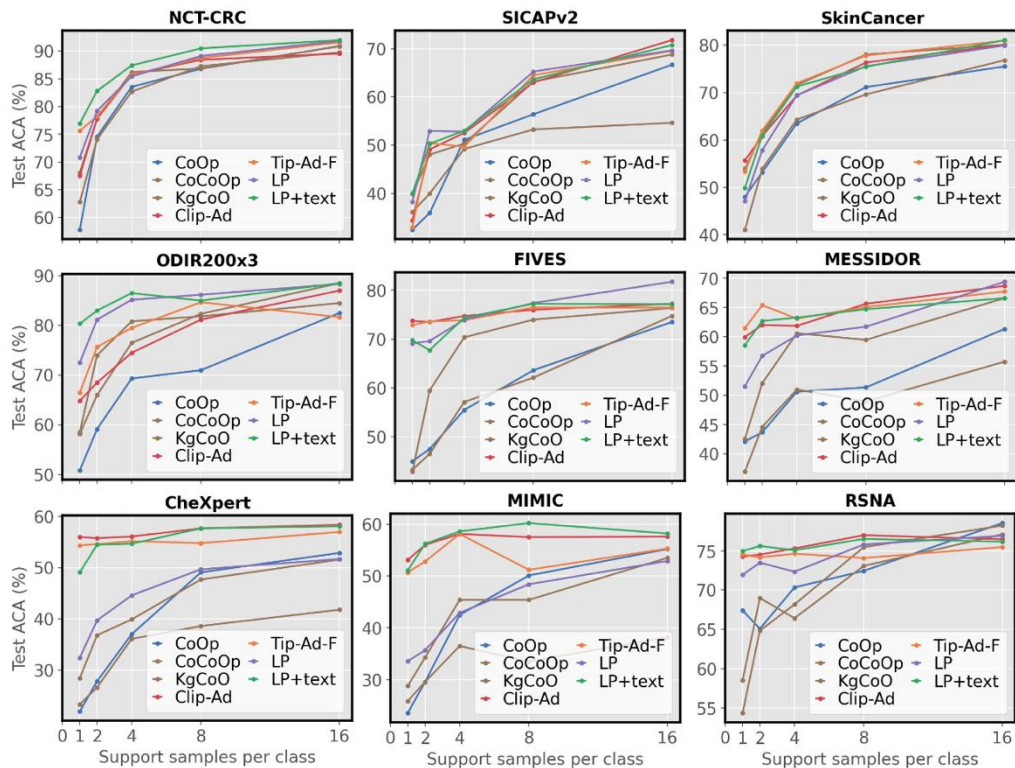


A structured benchmark for the few-shot adaptation of medical VLMs

- **Three** medical VLMs across different domains
- **Nine** tasks
- Benchmarking **Prompt Learning** and **Adapter-based** strategies
- and a strong **Linear Probe (LP++)**



A structured benchmark for the few-shot adaptation of medical VLMs



A structured benchmark for the few-shot adaptation of medical VLMs

Computational Efficiency. Experiments on a NVIDIA RTX A6000 GPU on NCT-CRC. $D_1 = 256$, and $D_2 = D = 512$. Number of context tokens for CoOp and KgCoOp: $n_{ctx1} = 16$; for CoCoOp: $n_{ctx2} = 4$.

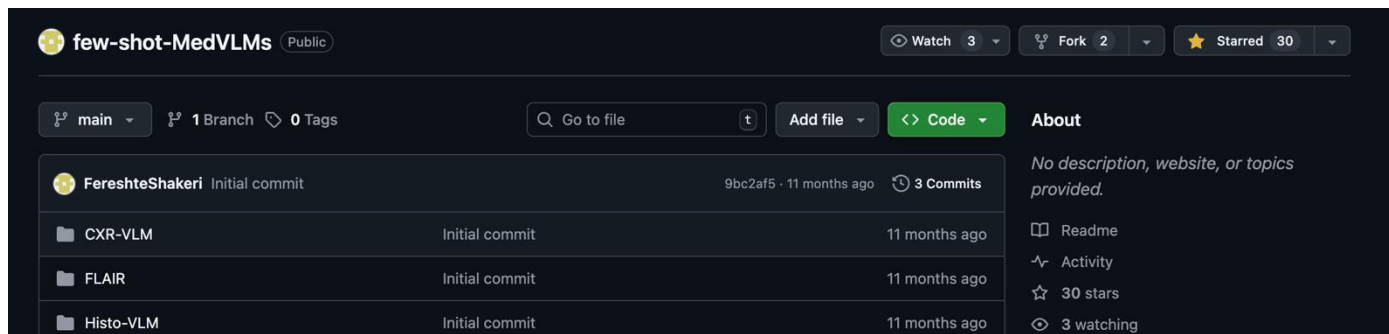
Methods	Category	Training Time	Black-box	#Parameters
Zero-shot		n/a	✓	n/a
CoOp [4]	<i>Prompt-Learning</i>	3min	✗	$K \times n_{ctx1} \times D$
CoCoOp [5]		12min	✗	$n_{ctx2} \times D + C$
KgCoOp [6]		3min	✗	$K \times n_{ctx1} \times D$
Clip-Adapter [7]	<i>CLIP-based Adapters</i>	2min	✓	$2(D_1 \times D_2)$
Tip-adapter-F [8]		2min	✓	$K \times S \times D$
LP	<i>Linear probe</i>	43s	✓	$K \times D$
LP+text [9]		4s	✓	$K(D + 1)$

GPU memory:

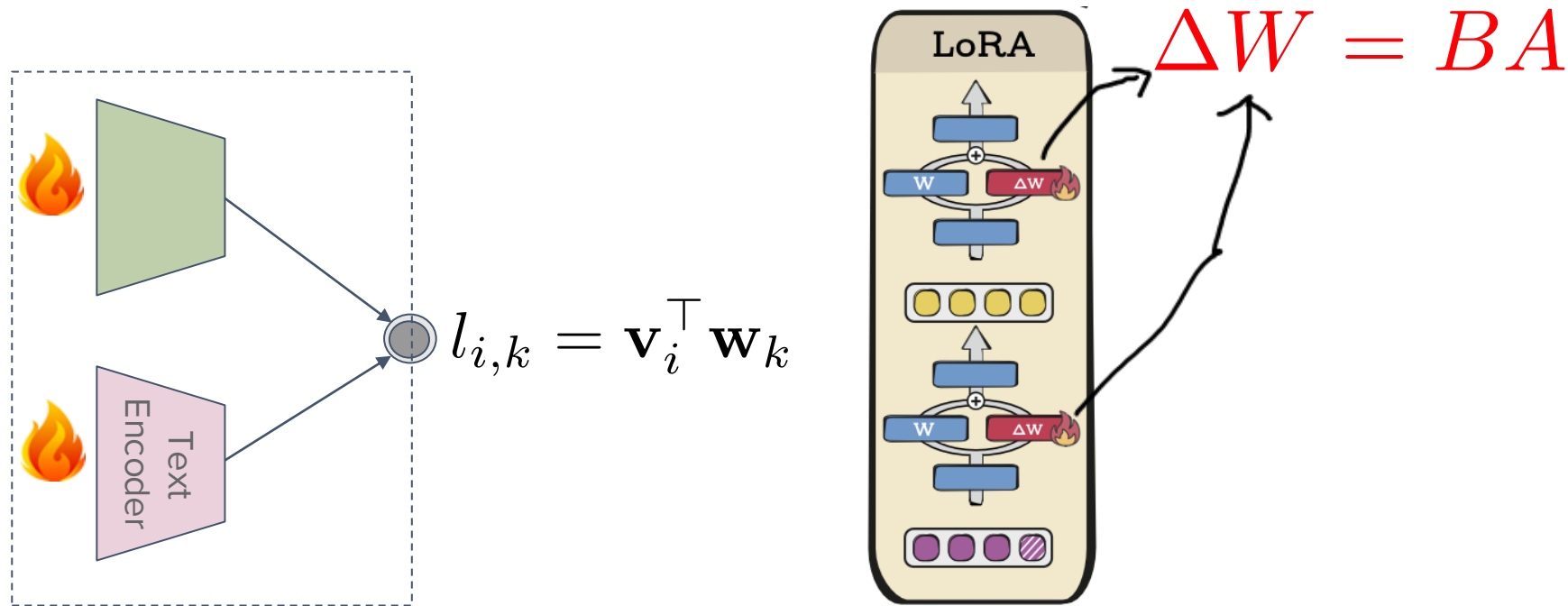
- LP+text: 800MB
- Prompt Learning: 28GB

The benchmark is publicly available

<https://github.com/FereshteShakeri/few-shot-MedVLMs>



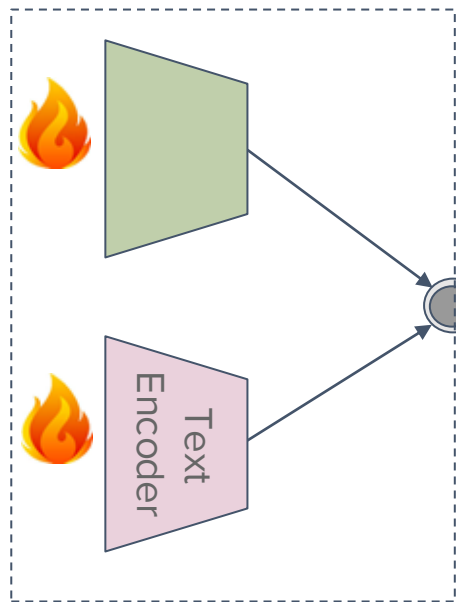
Low-rank adaptation for VLMs



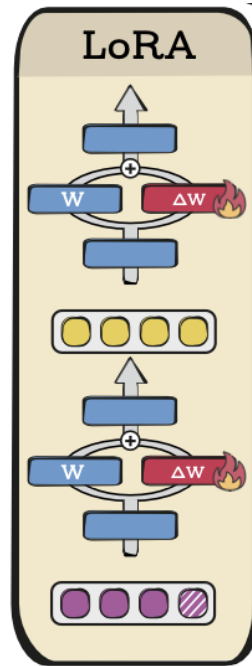
Zanella & Ben Ayed et al., Low-Rank Few-Shot Adaptation of Vision-Language Models, CVPRw 2024

Hu et al., LoRA: Low-Rank Adaptation of Large Language Models, ICLR 2022

Low-rank adaptation for VLMs



$$l_{i,k} = \mathbf{v}_i^\top \mathbf{w}_k$$



$$\Delta W = BA$$

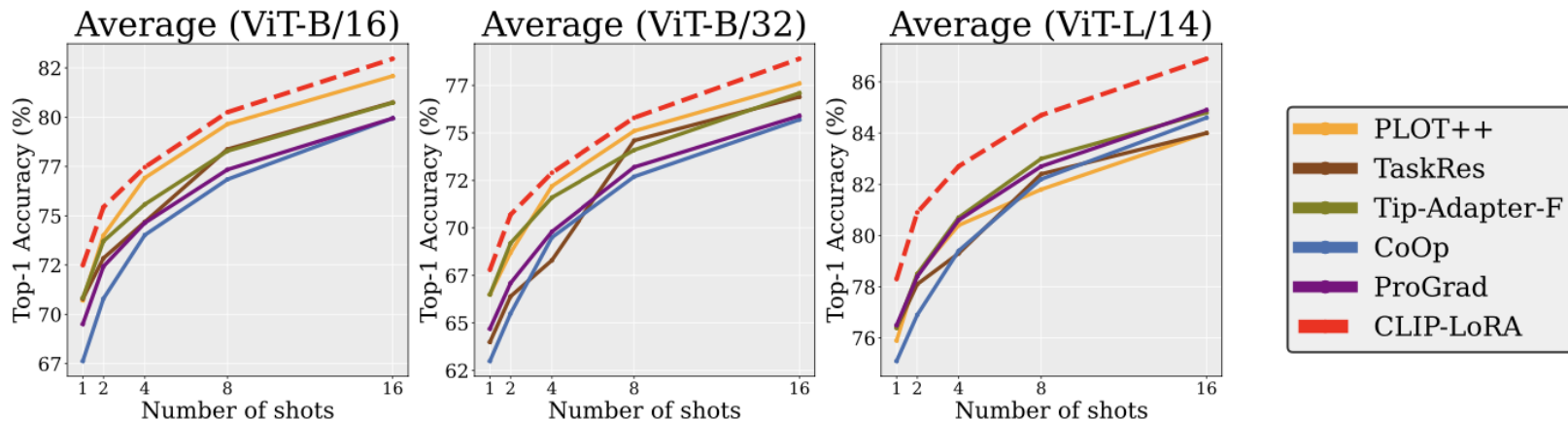
$\mathbb{R}^{m \times r}$ $\in \mathbb{R}^{r \times n}$

$$r \ll (m, n)$$

Zanella & Ben Ayed et al., Low-Rank Few-Shot Adaptation of Vision-Language Models, CVPRw 2024

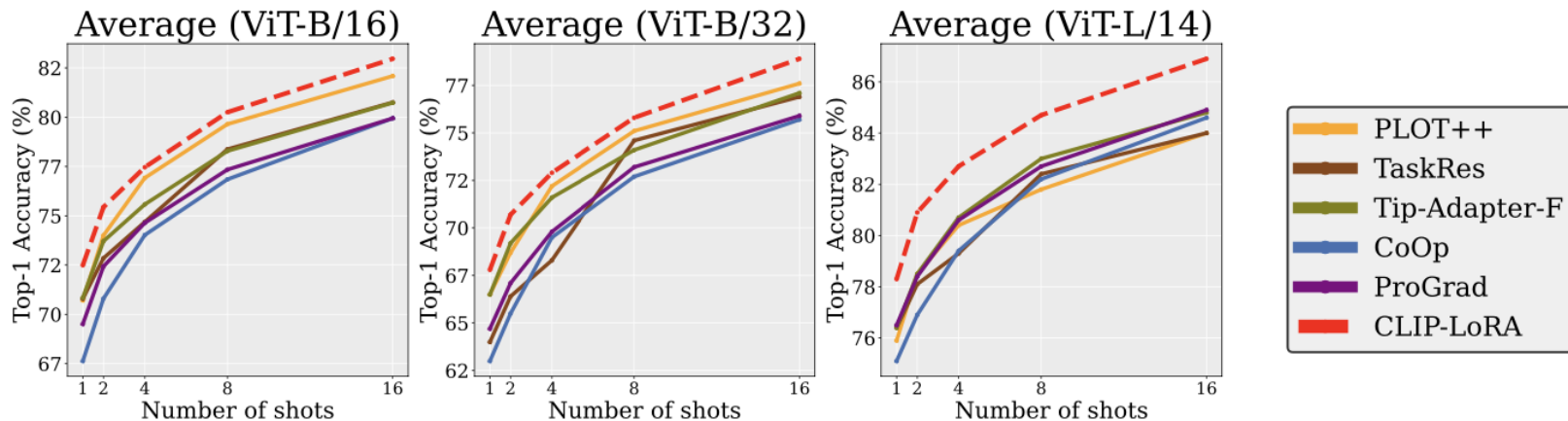
Hu et al., LoRA: Low-Rank Adaptation of Large Language Models, ICLR 2022

Low-rank adaptation of VLMs



Results from Zanella & Ben Ayed et al., Low-Rank Few-Shot Adaptation of Vision-Language Models, CVPRw 2024

Low-rank adaptation of VLMs



16-shot task on
one single A100 80Gb

Method	Training time
CoOp (16)	2h
PLOT++	15h30
ProGrad	3h20
CLIP-LoRA	50 min.

Code publicly available

<https://github.com/MaxZanella/CLIP-LoRA>

 **MaxZanella** / **CLIP-LoRA** Public

 Notifications  Fork 23  Star 227

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 main  1 Branch  0 Tags [Code](#)

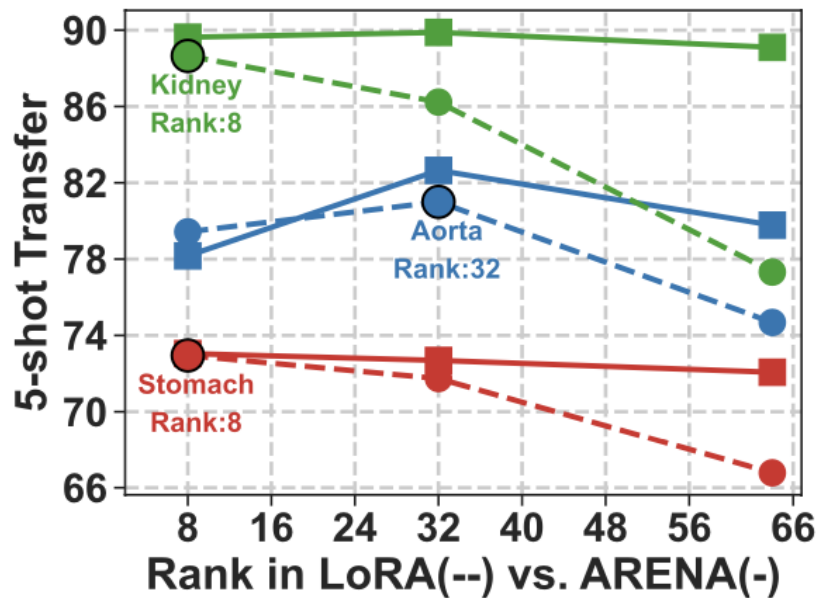
 **MaxZanella** Add files via upload ce3701b · last month  26 Commits

 clip	Initial commit	last year
 datasets	Initial commit	last year

About

An easy way to apply LoRA to CLIP. Implementation of the paper "Low-Rank Few-Shot Adaptation of Vision-Language Models" (CLIP-LoRA) [CVPRW 2024].

Regularized Low-Rank Adaptation



(a) Robustness

Background on the Low-Rank Approximation

$$W \in \mathbb{R}^{m \times n}$$

$$\min_{\hat{W}} \|W - \hat{W}\|_F \quad \text{s.t.} \quad \text{rank}(\hat{W}) \leq r$$


Background on the Low-Rank Approximation

$$W \in \mathbb{R}^{m \times n}$$

$$\min_{\hat{W}} \|W - \hat{W}\|_F \quad \text{s.t.} \quad \text{rank}(\hat{W}) \leq r$$

Eckart-Young-Mirsky Theorem

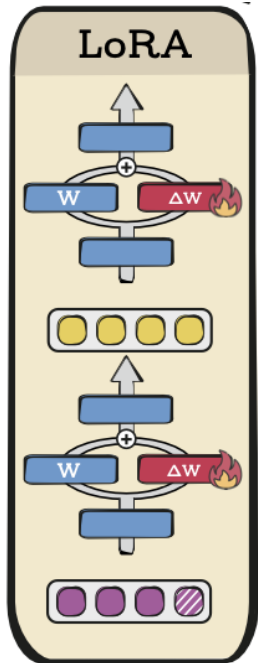
$$\hat{W} = P\Lambda Q^t$$

$$Q \in \mathbb{R}^{n \times r}$$

$$P \in \mathbb{R}^{m \times r}$$

Diagonal r by r containing the largest r singular values of W

Back to our problem



$$W = W_0 + \Delta W = W_0 + BA$$

$$A \in \mathbb{R}^{r \times n}$$

$$B \in \mathbb{R}^{m \times r}$$

$$r \ll (m, n)$$

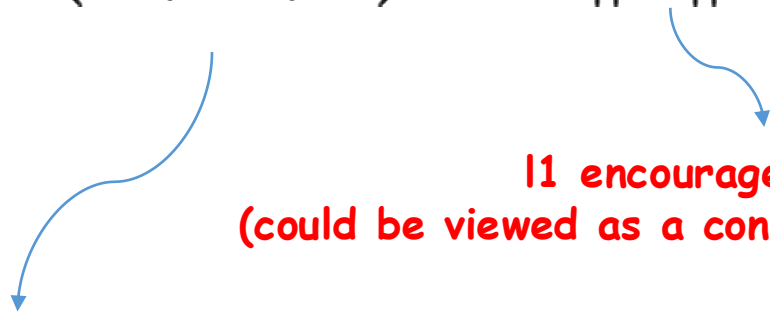
Intuition Behind Regularized Low-Rank Adaptation

$$W = W_0 + \Delta W = W_0 + B \text{Diag}(v) A$$

The number of non-zero elements of the vector of diagonal elements determine the rank of the decomposition


$$\|v\|_0$$

Regularized Low-Rank Adaptation

$$\mathcal{L}(A, B, v) + \lambda \|v\|_1$$
A diagram with two blue curved arrows. One arrow starts from the term $\mathcal{L}(A, B, v)$ and points down to the text 'Loss function of the task'. The other arrow starts from the term $\lambda \|v\|_1$ and points down to the text 'l1 encourages vector sparsity (could be viewed as a convex relaxation of the l0 norm)'.

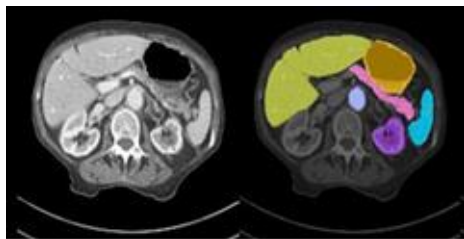
Loss function of the task

**l1 encourages vector sparsity
(could be viewed as a convex relaxation of the l0 norm)**

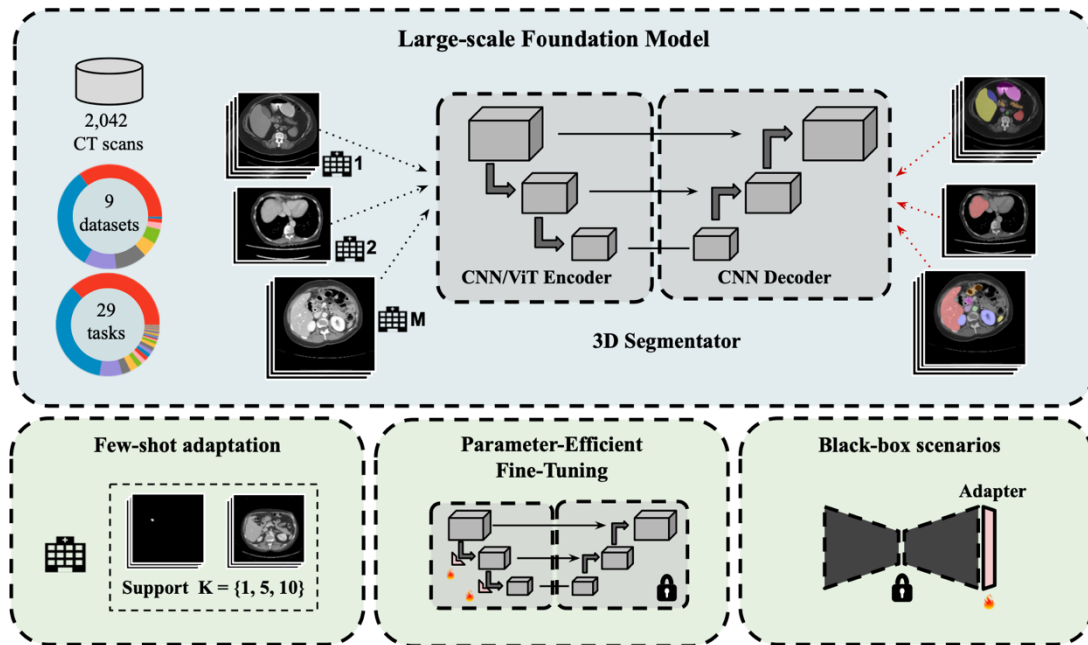
Experimental setup

Spleen Kidney Gallbladder Esophagus Liver Pancreas Stomach Duodenum Aorta

Training classes



Novel classes:
heart-myocardium
left atrium & right
atrium, left & right
ventricle



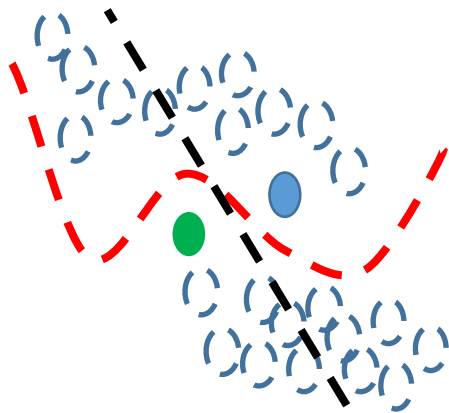
Some results: Transfer to novel organs

		Method	MYO	LA	RA	LV	RV	Avg.
10-shot		Linear probe	<u>64.50</u>	63.47	66.86	69.12	62.60	65.31
		BitFit [33]	64.18	64.15	66.35	<u>69.79</u>	62.61	64.42
		Affine-LN [1]	64.39	63.62	67.95	69.93	63.66	65.91
		FFT	59.07	54.05	63.06	64.38	59.50	60.01
		-----	-----	-----	-----	-----	-----	-----
		LoRA [15]	60.31	<u>65.2</u>	<u>78.44</u>	64.05	<u>65.29</u>	<u>66.66</u>
		ARENA (<i>Ours</i>)	75.29	81.8	82.93	74.2	74.82	77.81

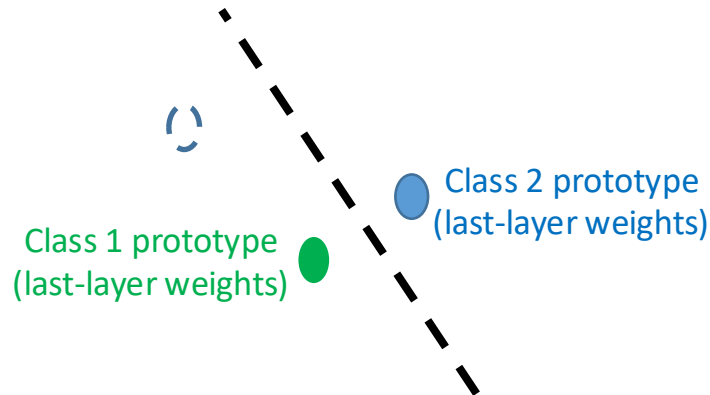
Some results: Transfer to training organs

		Method	Spl	lKid	Gall	Eso	Liv	Pan	Sto	Duo	Aor	Avg.
10-shot		Linear probe	91.72	89.78	78.49	47.01	92.16	78.14	76.80	63.63	69.91	76.40
		BitFit [33]	90.85	87.68	75.92	47.92	91.85	79.83	66.35	64.10	77.98	75.83
		Affine-LN [1]	<u>92.20</u>	86.02	79.58	50.27	89.98	77.64	<u>69.15</u>	<u>67.64</u>	<u>83.53</u>	77.33
		FFT	89.61	84.79	76.07	56.82	90.89	74.87	60.78	71.29	91.81	<u>77.44</u>
		LoRA [15]	89.94	89.47	<u>80.65</u>	46.11	<u>92.94</u>	81.18	66.41	61.76	81.66	76.68
		ARENA (<i>Ours</i>)	92.28	<u>89.58</u>	84.49	<u>50.83</u>	93.01	<u>80.27</u>	68.35	62.95	82.46	78.25

Transductive vs. inductive prediction

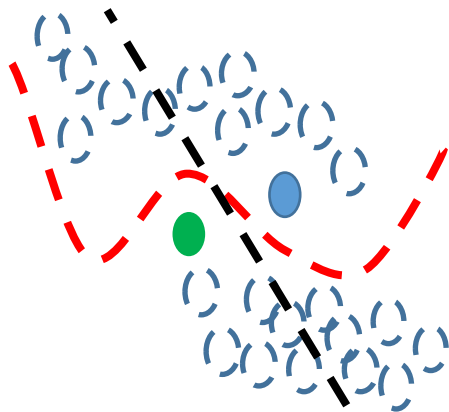


Transductive
(joint test-time prediction)



Inductive
(one test sample at a time)

Popular in few-shot learning (with standard vision models, not VLMs)



Transductive
(joint test-time prediction)

[Finn et al., ICML'17]
(**MAML: Transductive BatchNorm**)

[Dhillon et al., ICLR'20]
(**Entropy Minimization**)

[Ziko et al., ICML'20]
(**Laplacian Regularization**)

[Boudiaf et al., NeurIPS'20]
(**Information Maximization**)

Why transduction is a relevant and promising?

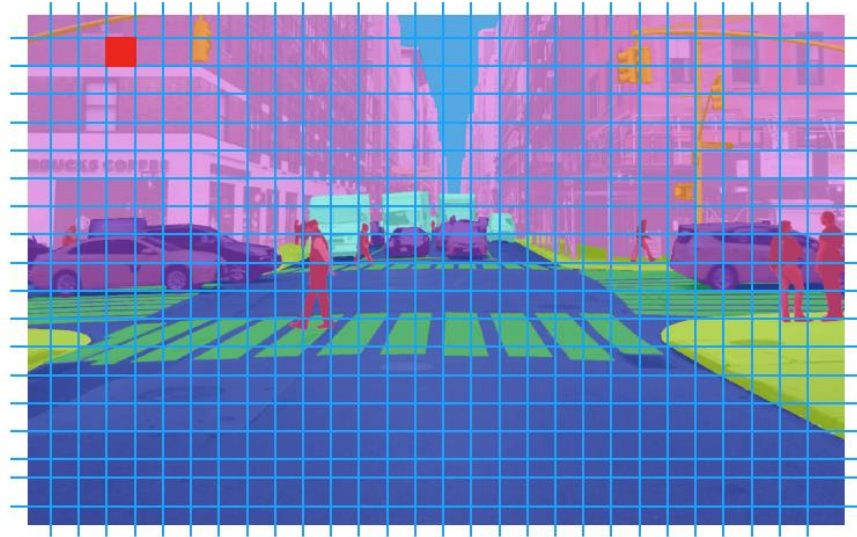
Practical scenarios: segmentation, videos streams, smart-device photos, autonomous vehicles, medical imaging, demographic or financial records, etc.



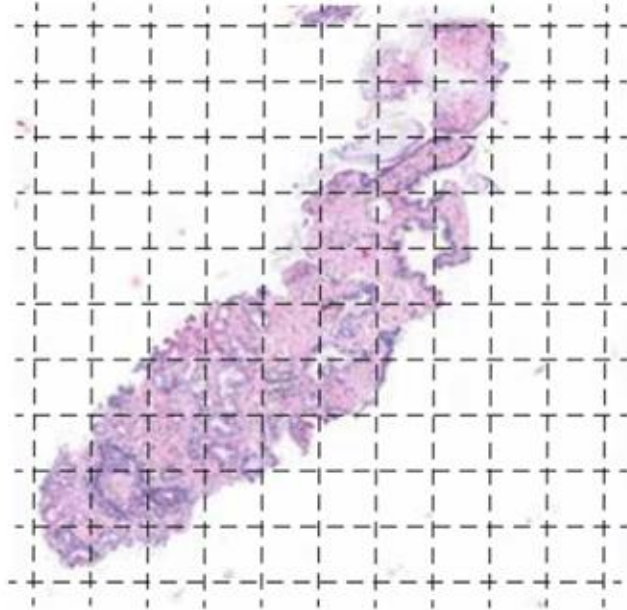
Why transduction is a relevant and promising?

Practical scenarios: segmentation, videos streams, smart-device photos, autonomous vehicles, medical imaging, demographic or financial records, etc.

Pixel = sample

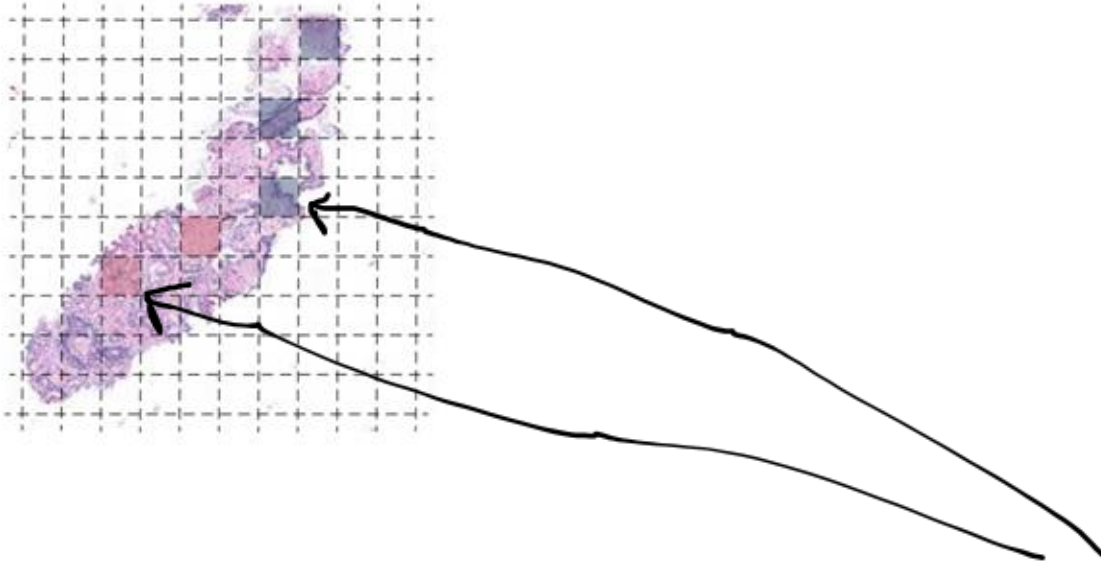


Why transduction is relevant



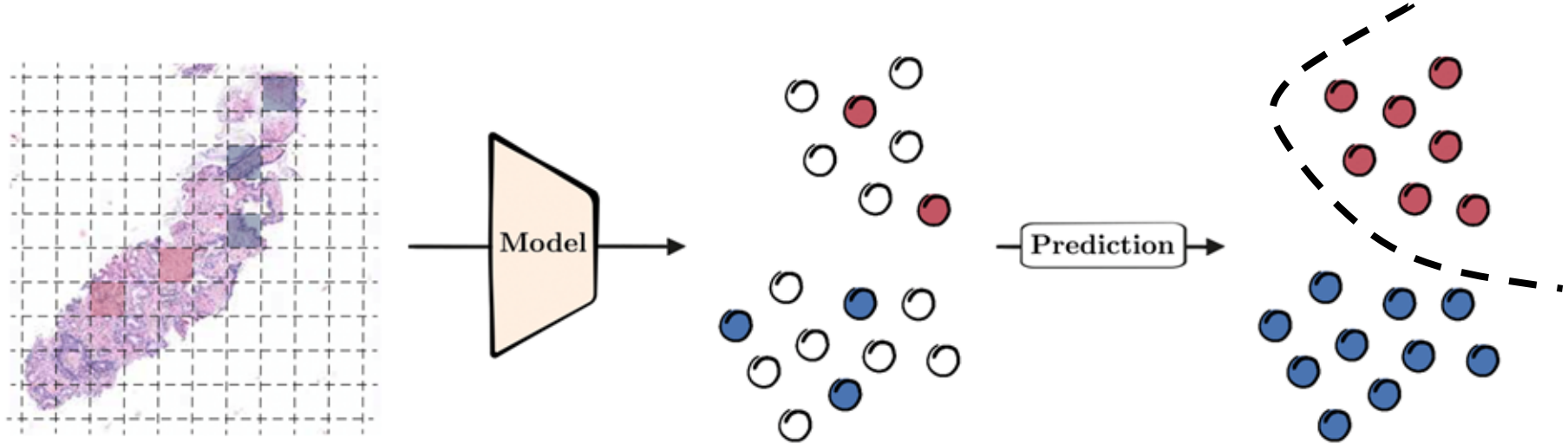
Example: Large whole-slide pathology images are very large and are typically divided into patches, each treated individually.

Standard transductive few-shot inference



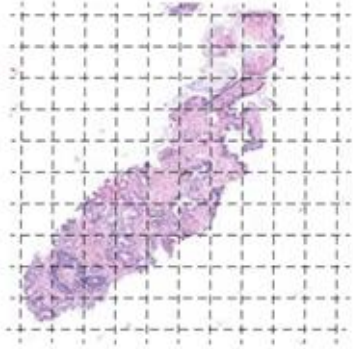
Large whole slides divided into patches + a few annotated patches

Standard transductive few-shot inference



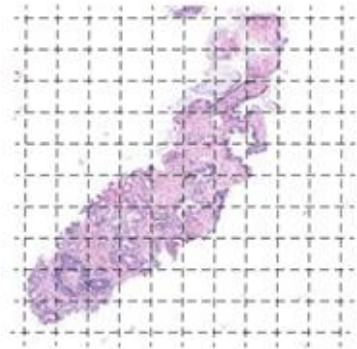
... we can leverage unlabeled **data statistics** & the few-labeled patches!

Unsupervised transductive inference (with VLMs)



Large whole slides divided into patches ~~+ a few annotated patches~~

Unsupervised transductive inference (with VLMs)

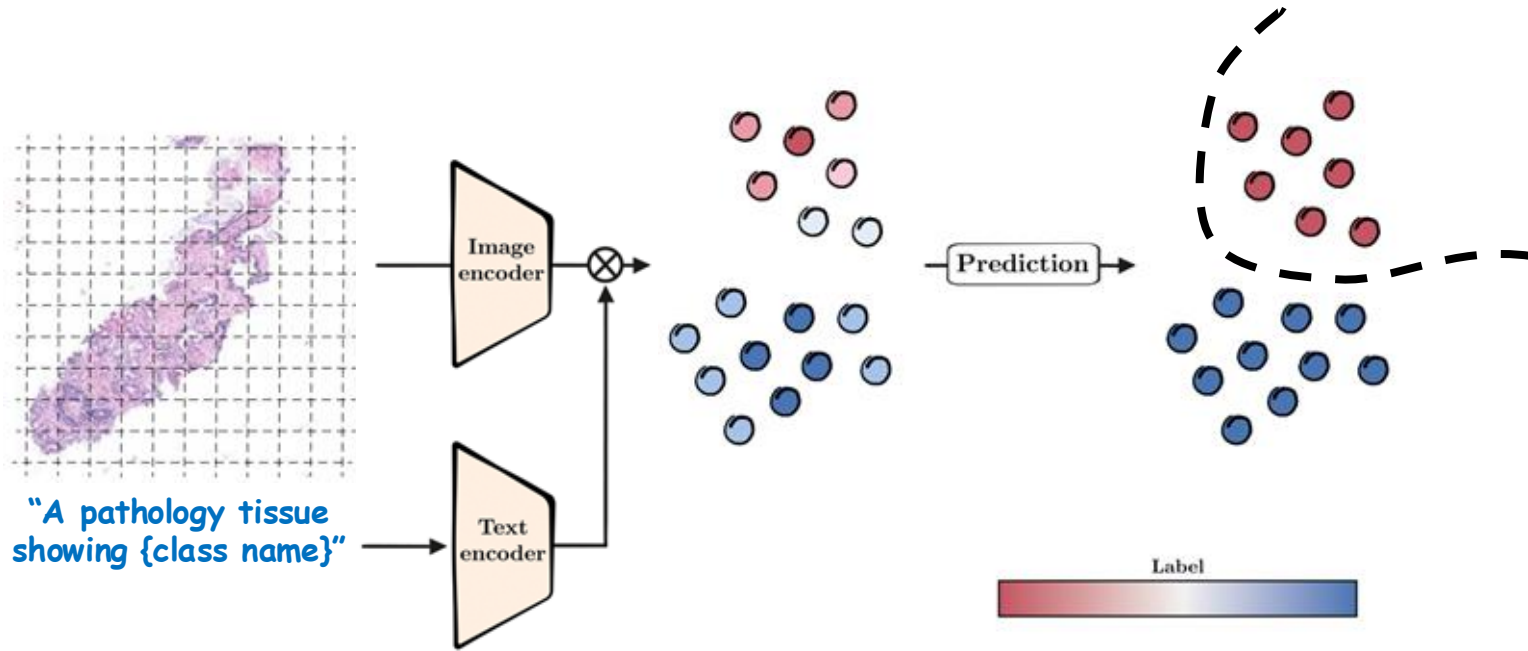


"A pathology tissue
showing {class name}"



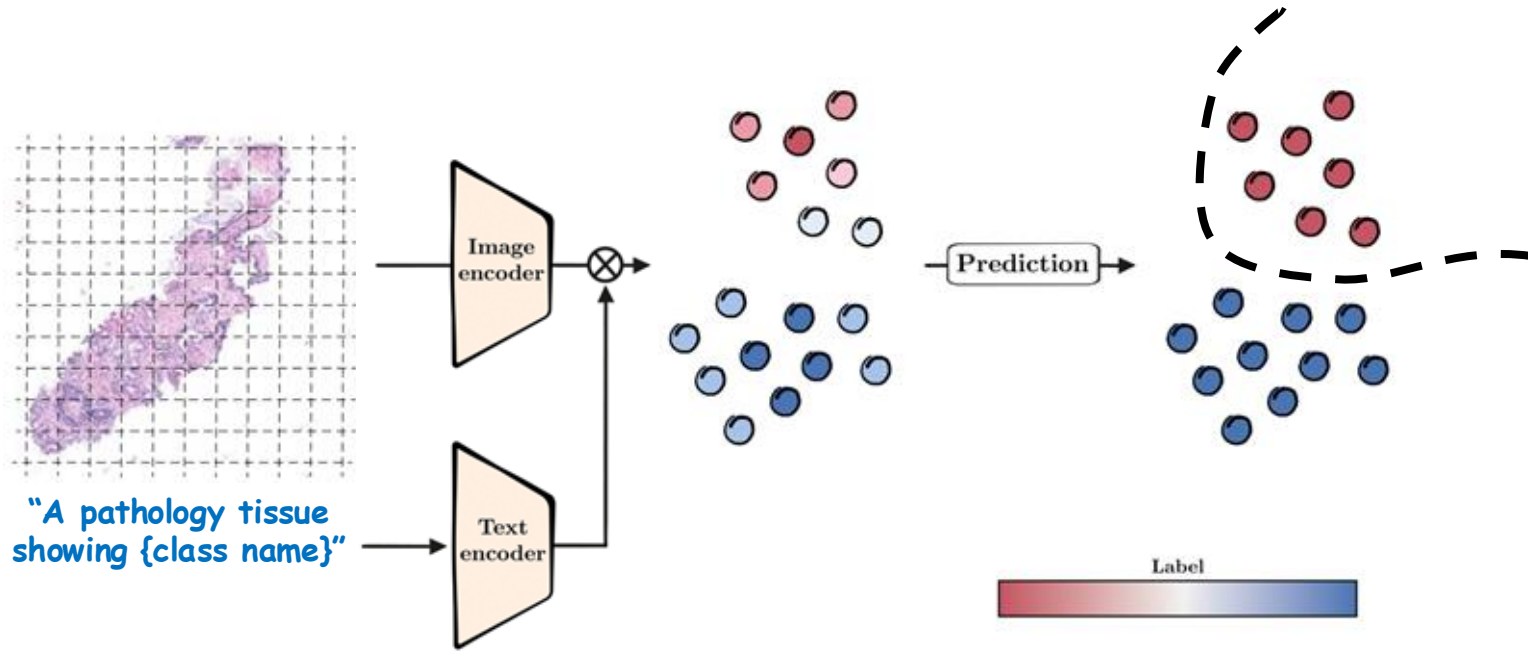
Large whole slides divided into patches + **text for each class**

Unsupervised transductive inference (with VLMs)



Text-image similarities give us (noisy) **zero-shot predictions** for each patch

Unsupervised transductive inference (with VLMs)

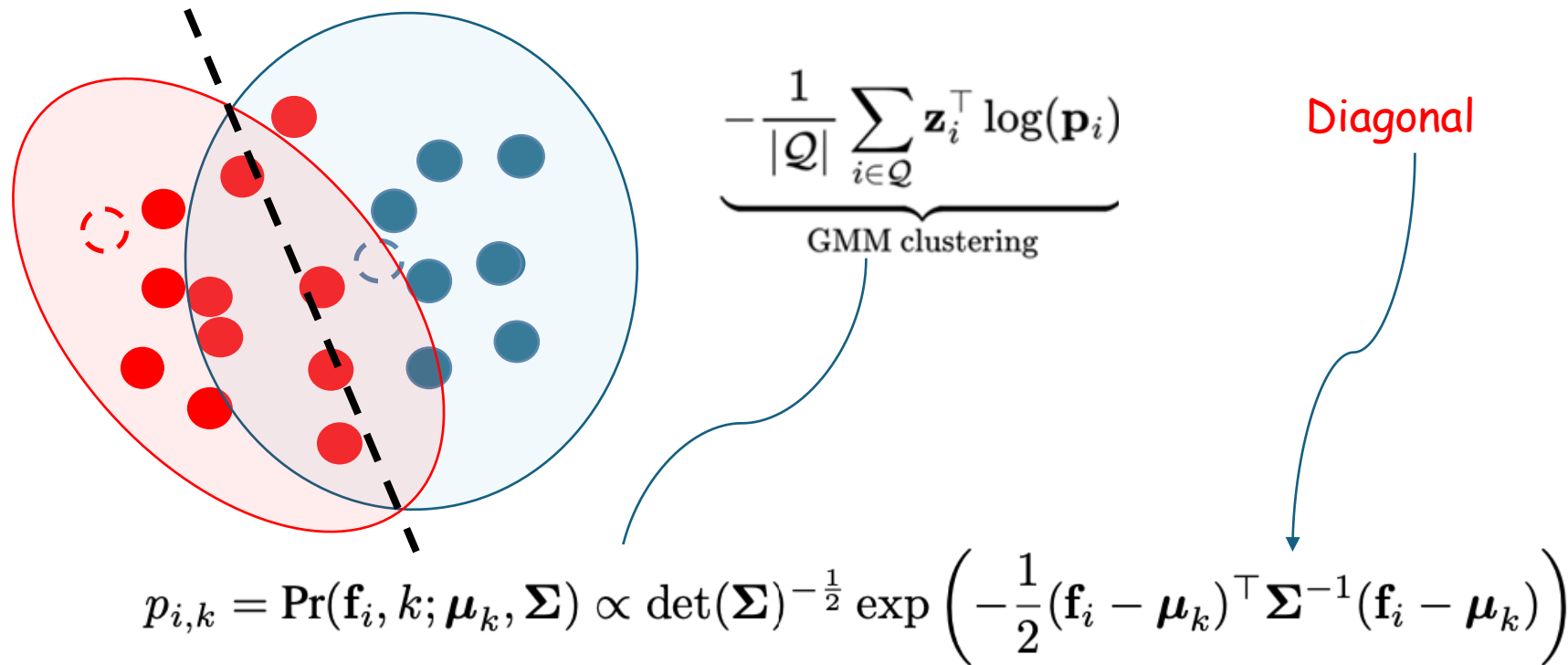


Also called test-time adaptation (in the context of VLMs)

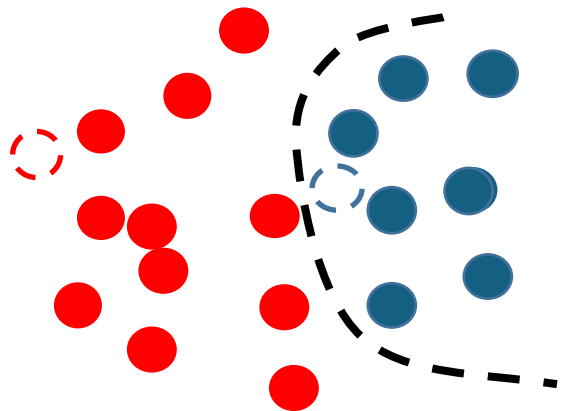
Regularized clustering for Few-shot/Zero-shot CLIP

$$\mathcal{L}_{\text{Zero-Shot}}(\mathbf{z}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \underbrace{-\frac{1}{|\mathcal{Q}|} \sum_{i \in \mathcal{Q}} \mathbf{z}_i^\top \log(\mathbf{p}_i)}_{\text{GMM clustering}} - \underbrace{\sum_{i \in \mathcal{D}} \sum_{j \in \mathcal{D}} w_{ij} \mathbf{z}_i \mathbf{z}_j}_{\text{Laplacian reg.}} + \underbrace{\sum_{i \in \mathcal{Q}} \text{KL}_\lambda(\mathbf{z}_i \| \hat{\mathbf{y}}_i)}_{\text{Text knowledge}}$$

Regularized clustering for Few-shot/Zero-shot CLIP



Regularized clustering for Few-shot/Zero-shot CLIP

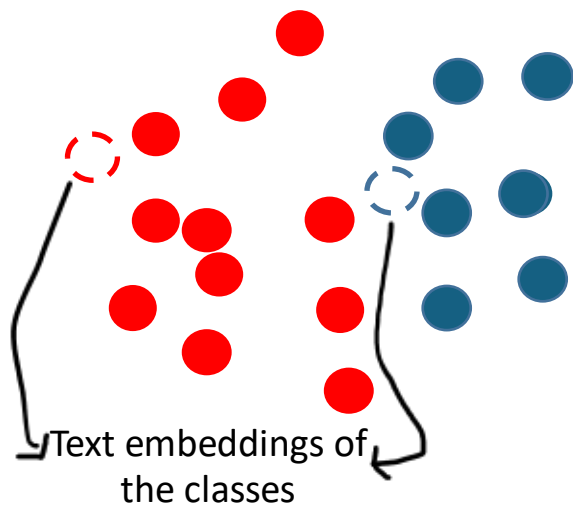


$$-\underbrace{\sum_{i \in \mathcal{D}} \sum_{j \in \mathcal{D}} w_{ij} \mathbf{z}_i \mathbf{z}_j}_{\text{Laplacian reg.}}$$

$$w_{ij} = \mathbf{f}_i^\top \mathbf{f}_j$$

Gram
matrix

Regularized clustering for Few-shot/Zero-shot CLIP



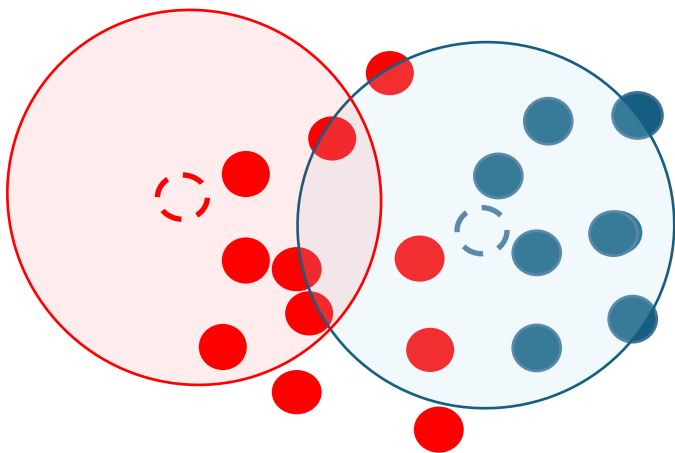
$$\underbrace{\sum_{i \in \mathcal{Q}} \text{KL}_\lambda(\mathbf{z}_i || \hat{\mathbf{y}}_i)}_{\text{Text knowledge}}$$

$$\text{KL}_\lambda(\mathbf{z}_i || \hat{\mathbf{y}}_i) = \mathbf{z}_i^\top \log \mathbf{z}_i - \lambda \mathbf{z}_i^\top \log \hat{\mathbf{y}}_i, \quad i \in \mathcal{Q}; \quad \lambda > 0$$

$$\hat{\mathbf{y}}_i = (\hat{y}_{i,k})_{1 \leq k \leq K} \in \Delta_K; \quad \hat{y}_{i,k} = \frac{\exp(\tau \mathbf{f}_i^\top \mathbf{t}_k)}{\sum_j \exp(\tau \mathbf{f}_i^\top \mathbf{t}_j)}$$

To summarize TransCLIP

$$\mathcal{L}_{\text{Zero-Shot}}(\mathbf{z}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \underbrace{-\frac{1}{|\mathcal{Q}|} \sum_{i \in \mathcal{Q}} \mathbf{z}_i^\top \log(\mathbf{p}_i)}_{\text{GMM clustering}} - \underbrace{\sum_{i \in \mathcal{D}} \sum_{j \in \mathcal{D}} w_{ij} \mathbf{z}_i \mathbf{z}_j}_{\text{Laplacian reg.}} + \underbrace{\sum_{i \in \mathcal{Q}} \text{KL}_\lambda(\mathbf{z}_i || \hat{\mathbf{y}}_i)}_{\text{Text knowledge}}$$

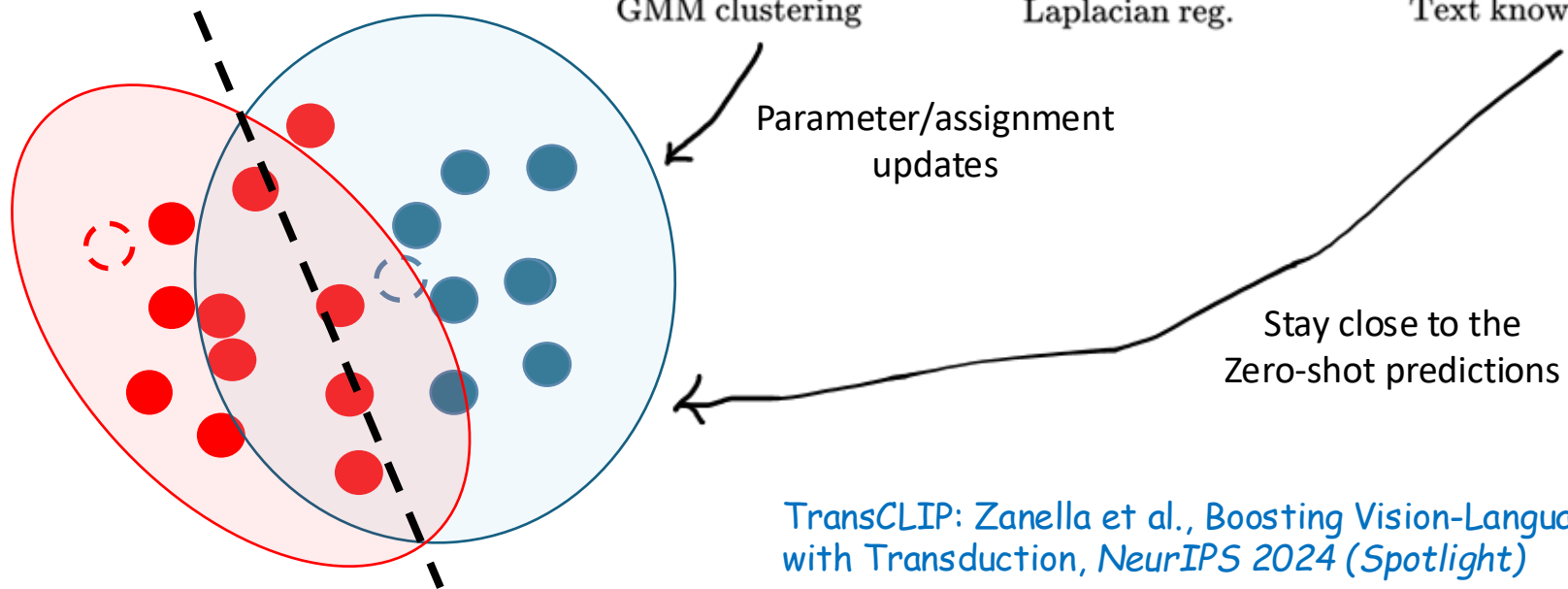


Initialization

TransCLIP: Zanella et al., Boosting Vision-Language Models with Transduction, *NeurIPS 2024 (Spotlight)*

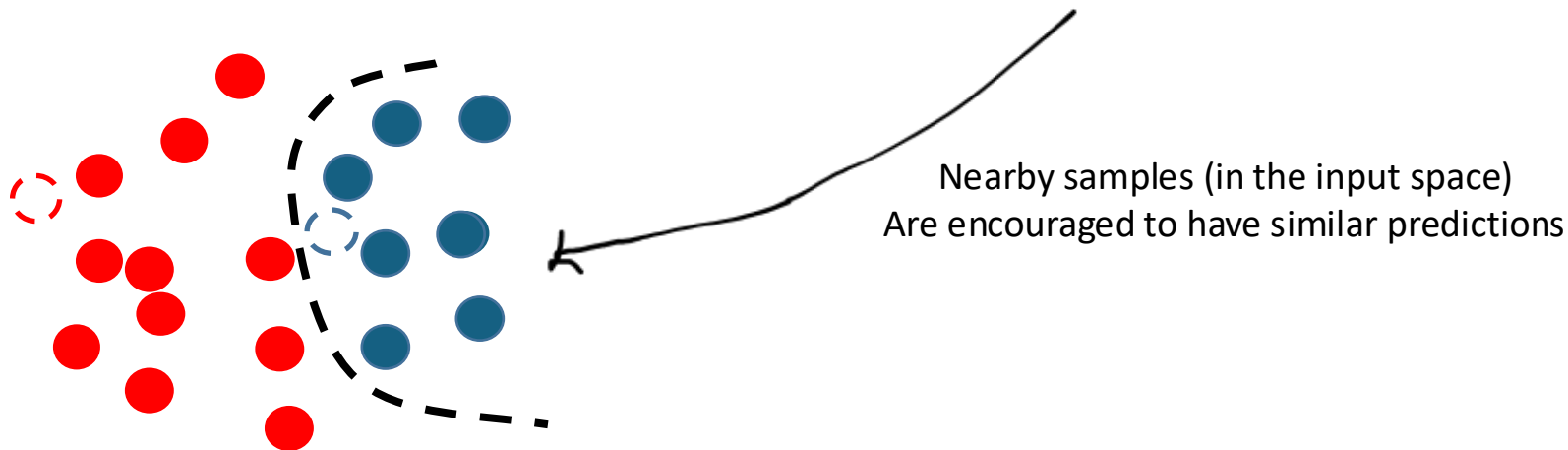
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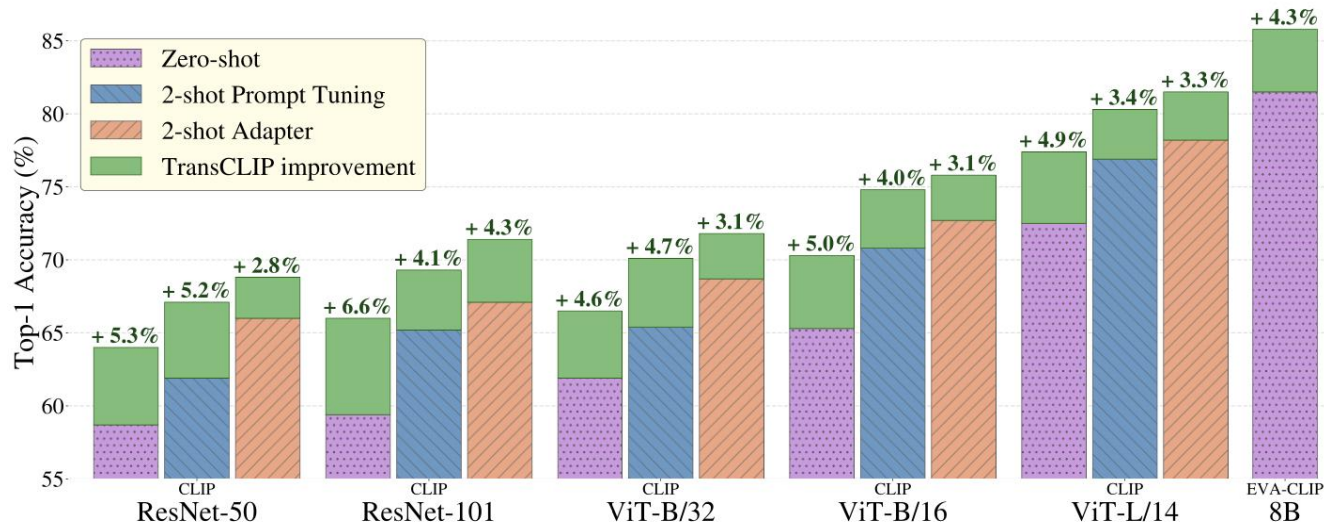
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TransCLIP: Zanella et al., Boosting Vision-Language Models with Transduction, *NeurIPS 2024 (Spotlight)*

TransCLIP's boost

On 11 natural image
datasets



TransCLIP

Procedure of 3 update equations
→ a few lines of code!

#Patches	Features	Histo-TransCLIP
10^2	~ 1 sec.	~ 0.1 sec.
10^3	~ 4 sec.	~ 0.2 sec.
10^4	~ 28 sec.	~ 0.4 sec.
10^5	~ 5 min.	~ 6 sec.

```
1 function Histo-TransCLIP(f, t,  $\tau$ )  
  // Text-based pseudo-labels  $\hat{y}$   
2  $\hat{y}_i = \text{softmax}(\tau \mathbf{f}_i^T \mathbf{t}) \quad \forall i$   
  // Initialize  $\mathbf{z}$ ,  $\boldsymbol{\mu}$ ,  $\boldsymbol{\Sigma}$   
3  $\mathbf{z}_i = \hat{y}_i \quad \forall i$   
4  $\boldsymbol{\mu}_k = \text{top\_confident\_average}(\mathbf{f}, \hat{y}) \quad \forall k$   
5  $\text{diag}(\boldsymbol{\Sigma}) = \frac{1}{n\_features}$   
  // Iterative procedure  
6 while not_converged do  
7   for  $l = 1 : \dots$  do  
8      $\mathbf{z}_i^{(l+1)} = \frac{\hat{y}_i \odot \exp(\log(\mathbf{p}_i) + \sum_{j \in Q} w_{ij} \mathbf{z}_j^{(l)})}{(\hat{y}_i \odot \exp(\log(\mathbf{p}_i) + \sum_{j \in Q} w_{ij} \mathbf{z}_j^{(l)}))^T \mathbf{1}_K} \quad \forall i$   
9      $\boldsymbol{\mu}_k = \frac{\sum_{i \in Q} z_{i,k} \mathbf{f}_i}{\sum_{i \in Q} z_{i,k}} \quad \forall k$   
10     $\text{diag}(\boldsymbol{\Sigma}) = \frac{1}{|Q|} \sum_{i \in Q} \sum_k z_{i,k} (\mathbf{f}_i - \boldsymbol{\mu}_k)^2$   
11  return  $\mathbf{z}$ 
```

TransCLIP

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```

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7   for l = 1:... do
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11  return  $\mathbf{z}$ 

```

Dataset	#samples		Images + Texts encoding	Total	Top-1 accuracy
ImageNet	50,000	CLIP	58.7 sec	58.7 sec	66.6
		UPL*	58.7 sec	152 min	69.6
		TransCLIP-ZS	58.7 sec	73.1 sec	70.3

TransCLIP results on histology

Dataset	Method	Model				
		CLIP	Quilt-B16	Quilt-B32	PLIP	CONCH
<i>SICAP-MIL</i>	Zero-shot	29.85	40.44	35.04	46.84	27.71
	Histo-TransCLIP	24.72	58.49	28.18	53.23	32.58
<i>LC(Lung)</i>	Zero-shot	31.46	43.00	76.24	84.96	84.81
	Histo-TransCLIP	25.62	50.53	93.93	93.80	96.29
<i>SKINCANCER</i>	Zero-shot	4.20	15.38	39.71	22.90	58.53
	Histo-TransCLIP	11.46	33.33	48.80	36.72	66.22
<i>NCT-CRC</i>	Zero-shot	25.39	29.61	53.73	63.17	66.27
	Histo-TransCLIP	39.61	48.40	58.13	77.53	70.36
<i>Average</i>	Zero-shot	22.73	32.1	51.18	54.47	59.33
	Histo-TransCLIP	25.35	47.69	57.26	65.32	66.36
	$\Delta_{\text{transductive}}$	+2.62	+15.59	+6.08	+10.85	+7.03

Zanella et al., Boosting Vision-Language Models for Histopathology Classification: Predict all at once, Second International Workshop on Foundation Models for General Medical AI, MedAGI 2024

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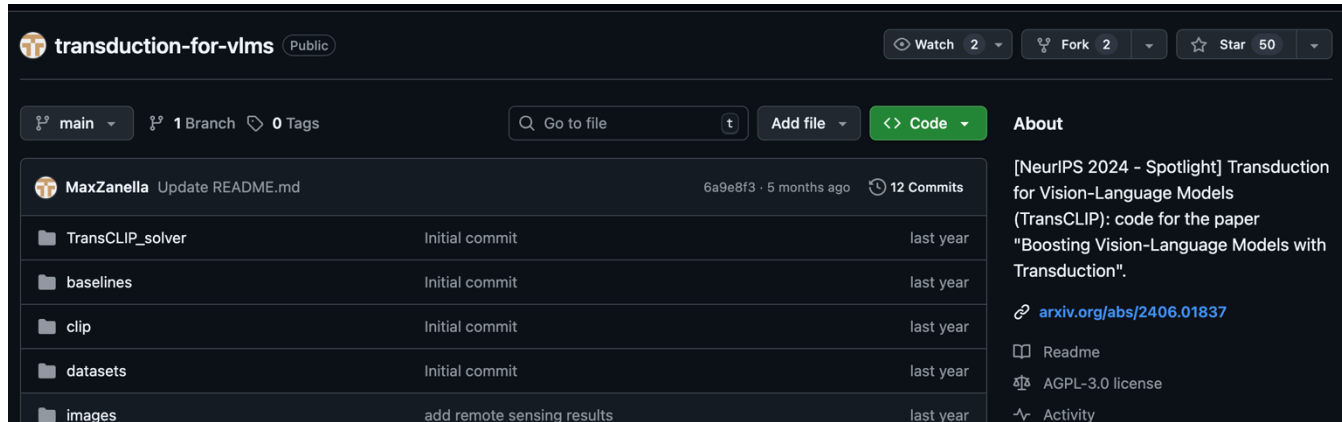
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Code publicly available

<https://github.com/MaxZanella/transduction-for-vlms>



The screenshot shows the GitHub repository page for 'transduction-for-vlms' by MaxZanella. The repository is public and has 2 watchers, 2 forks, and 50 stars. It is currently on the 'main' branch with 1 branch and 0 tags. The repository was last updated 5 months ago with commit 6a9e8f3. It has 12 commits. The repository contains the following files and folders:

File/Folder	Commit Message	Commit Date
TransCLIP_solver	Initial commit	last year
baselines	Initial commit	last year
clip	Initial commit	last year
datasets	Initial commit	last year
images	add remote sensing results	last year

The 'About' section on the right provides more details about the repository:

- [NeurIPS 2024 - Spotlight] Transduction for Vision-Language Models (TransCLIP): code for the paper "Boosting Vision-Language Models with Transduction".
- Link to the paper: arxiv.org/abs/2406.01837
- Readme
- AGPL-3.0 license
- Activity

TransCLIP: Zanella et al., Boosting Vision-Language Models with Transduction, *NeurIPS 2024 (Spotlight)*

To summarize...

- VLMs are very promising in medical imaging

If done properly: Specialized, knowledge-aware, discriminative when it matters

- Fine-tuning and adaptation:

Not resolved: What and how to finetune, model selection, doing it locally...

- Interpretability and confidence calibration are keys for wider adoption